ppt em baixo

RIP

180s + 30s (x nós do caminho)

5OGbit ethernet

Custos OSPF $= \dfrac{10^8}{\text{custo da rota}}$

Mudar custo da porta à mão
(pq eram todas de custo 1)

Área de backbone (de interligação) ⟶ área 0 ⟶ todos os routers têm que passar por ela

OSPF faz load-balance (balancear a utilização da carga)
↳ Redundância Ativa

Spanning-Tree ⟶ Redundância não-ativa

Router ID

↳ Maior endereço de loopback (sempre ativas)
↳ " " de uma porta física (não estão sempre ativas)

P/ condicionar o Router ID dos routers
Pode ser alterado pelo administrador

Não posso ter $\overbrace{\text{o mesmo}}^{\text{vários}}$ Router ID na mesma área

Routers de fronteira

10.1.0.0 /16
10.0.0.0 /16 } /15 Sumarização de rotas para o encaminhamento e anúncio de rotas

default-information originate 0.0.0.0 por dhcp

Túneis

Stub Areas

Planeamento de Redes-18013

Apresentação 6 – Routing interno

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Sumário

- Revisão do conceito de sistema autónomo
- Encaminhamento interno
  - Revisão acerca do encaminhamento RIPv1 e RIPv2
  - Revisão dos conceitos básicos acerca de encaminhamento OSPF
  - Configuração de OSPF por rede e por interface
  - Encaminhamento OSPF hierárquico:
    - Conceito de área
    - Tipos de área existentes
    - Mecanismos de interligação entre áreas OSPF
  - Rotas OSPF internas e externas
  - Outros protocolos de encaminhamento interno

Só para lembrar

Routing Tables

- An IP address may have multiple matches on a Routing Table:
  - Example: 192.168.1.12
  - Will match:
    - – 192.168.1.0/25via...
    - – 192.168.1.0/24via... – 192.168.0.0/23via... – 192.168.0.0/16via... –...
- Router will choose entry with the largest network prefix (most specific network).
  - i.e.,192.168.1.0/25via...

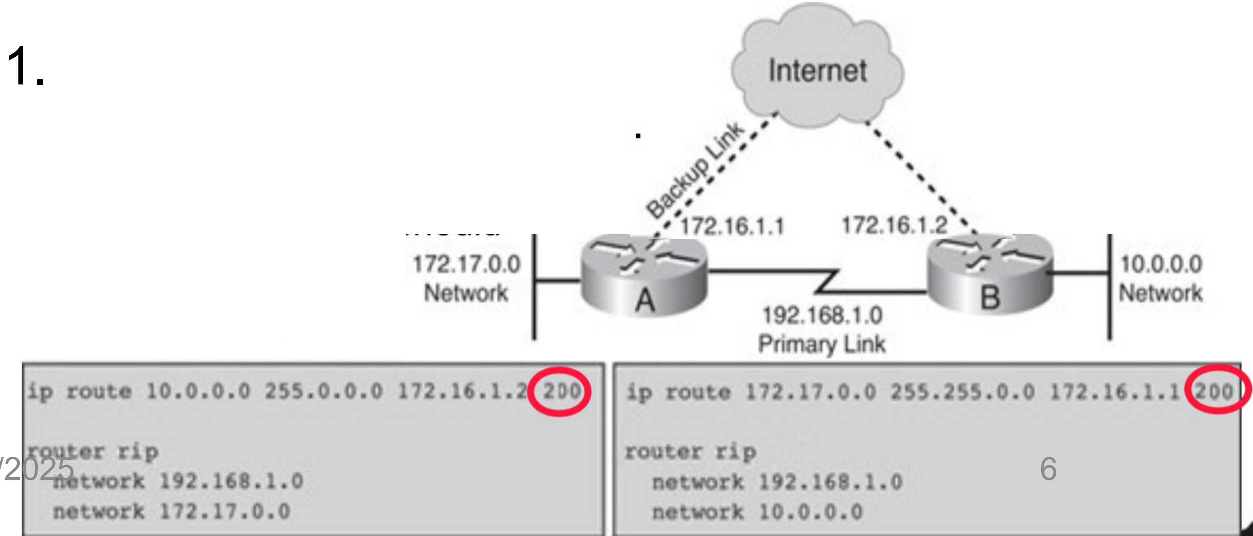
- Load balancing
  - Routing tables may have more than one path for each network
  - Traffic will be divided by all entries.
  - By packet, flow (TCP session, UDP IPs/port), etc...
    - – E.g, packet 1 path 1, packet 2 path 2, packet 3 path 1, ...
    - – Flow 1 path 1, flow 2 path 2, flow 3 path 3, flow 4 path 1, flow 5 path 2, ...

Administrative distance

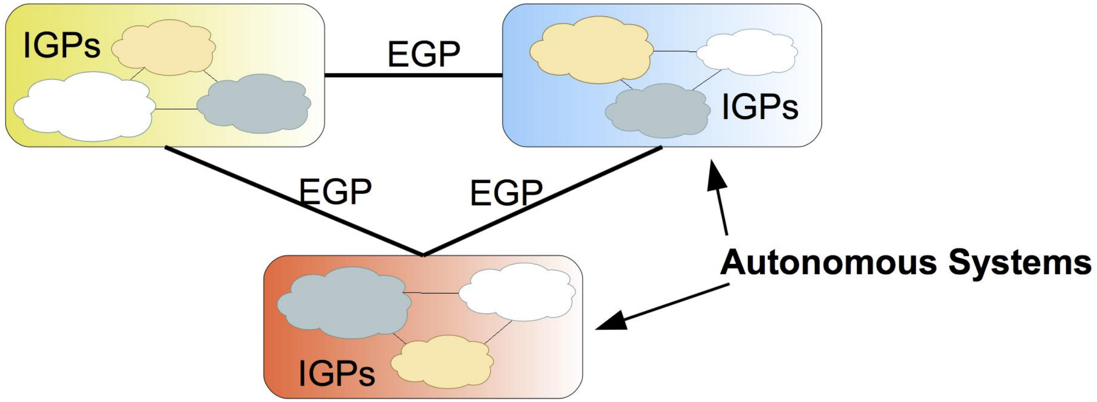
- Most routing protocols have metric structures and algorithms that are incompatible with other protocols.
- It is critical that a network using multiple routing protocols be able to seamlessly exchange route information and be able to select the best path across multiple protocols.
- Routers use a value called administrative distance to select the best path when they learn from different routing protocols the same destination (same network prefix and mask length).
- The Protocol/Method with the lowest Administrative Distance is preferred The Administrative Distance value is configurable.
- Example:
  - Static [1/1] 192.168.1.0/24 via ... ← Chosen!
  - RIP [120/1] 192.168.1.0/24 via ...
  - OSPF [110/1] 192.168.1.0/24 via ...

Floating Static Routes

- Based on the default administrative distances, routers use static routes over any dynamically learned route.
  - However, this default behavior might not be the desired behavior.
  - For example, when you configure a static route as a backup to a dynamically learned route, you do not want the static route to be used as long as the dynamic route is available.
- A static route that appears in the routing table only when the primary route goes away is called a floating static route.
  - The administrative distance of the static route is configured to be higher than the administrative distance of the primary route and it "floats" above the primary route, until the primary route is no longer available.
- RIP default administrative distance is 120.
- Static Routes default administrative distance is 1.
  - To create a floating static route (to backup a
  - RIP route) the administrative distance should
  - be greater than 120.
    - In example: 200.

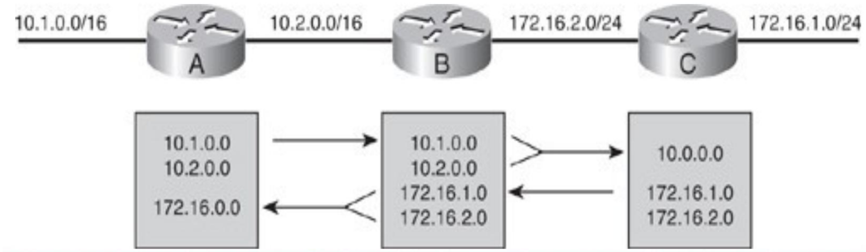


Autonomous System



- AS (Autonomous System) – set of routers/networks with a common routing policy and under the same administration.
- Routing inside an AS is performed by IGPs (Interior Gateway Protocols) such as RIPv1, RIPv2, OSPF, IS-IS and EIGRP.
  - Called Internal Routing
- Routing between AS is performed by EGPs (Exterior Gateway Protocols) such as BGP.
- IGPs and EGPs have different objectives:
  - IGPs: optimize routing performance
  - EGPs: optimize routing performance obeying political, economic and security policies.

Classful Routing Protocol Behavior



- A classful routing protocol does not include subnet mask information in its routing updates.
  - When a classful router receives routing updates, the router makes assumptions about the subnet mask being used by the networks listed in the update, based on IP address class.
  - A router sends the entire subnet address in the update,
  - The receiving router then assumes that the mask of the subnet in the update is the same as the mask on the receiving interface.

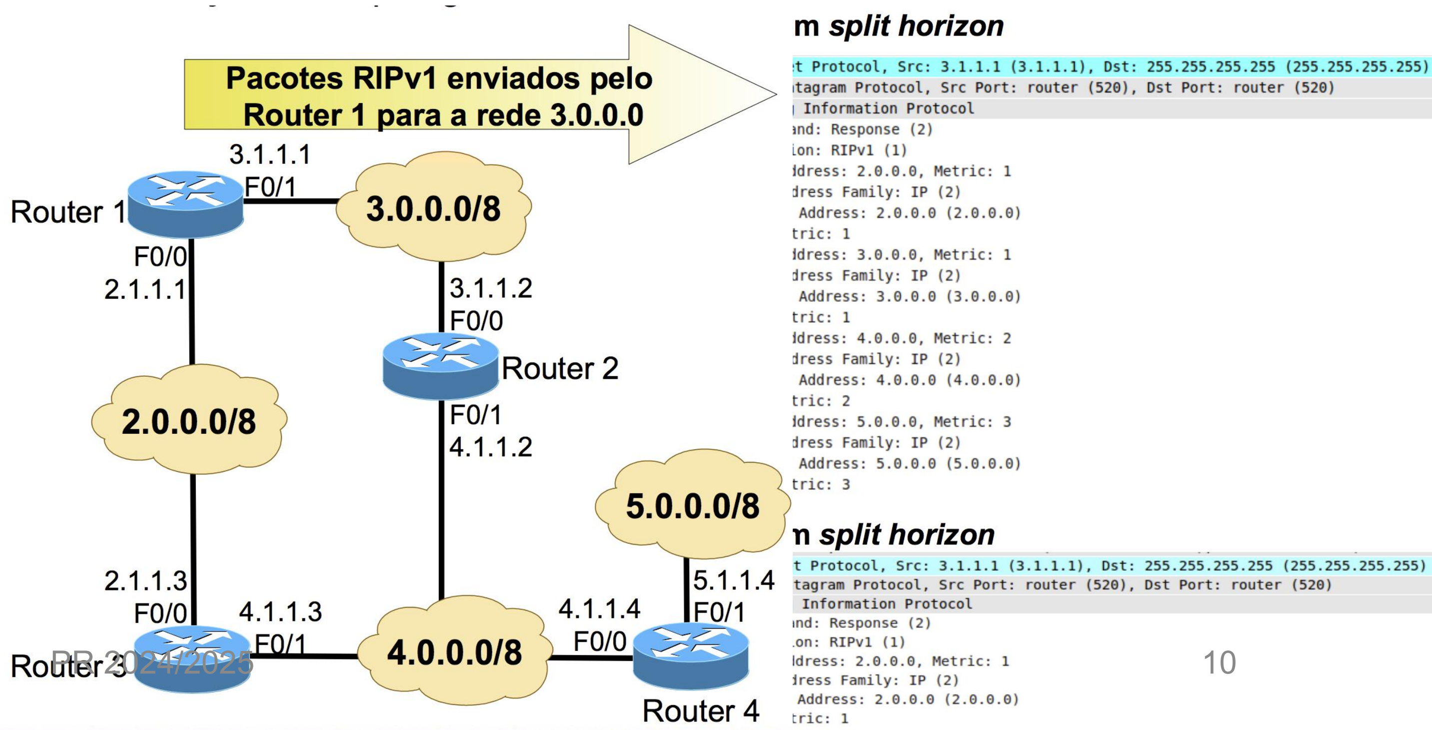
- When a router that is using a classful routing protocol needs to send an update about a subnet of a network across an interface belonging to a different network,
  - Assumes that the remote router will use the default subnet mask for that class of IP address.
  - It does not include the subnet information, the update packet contains only the major (classful) network information.
  - This process is called auto summarization across the network boundary.

RIP Version 1

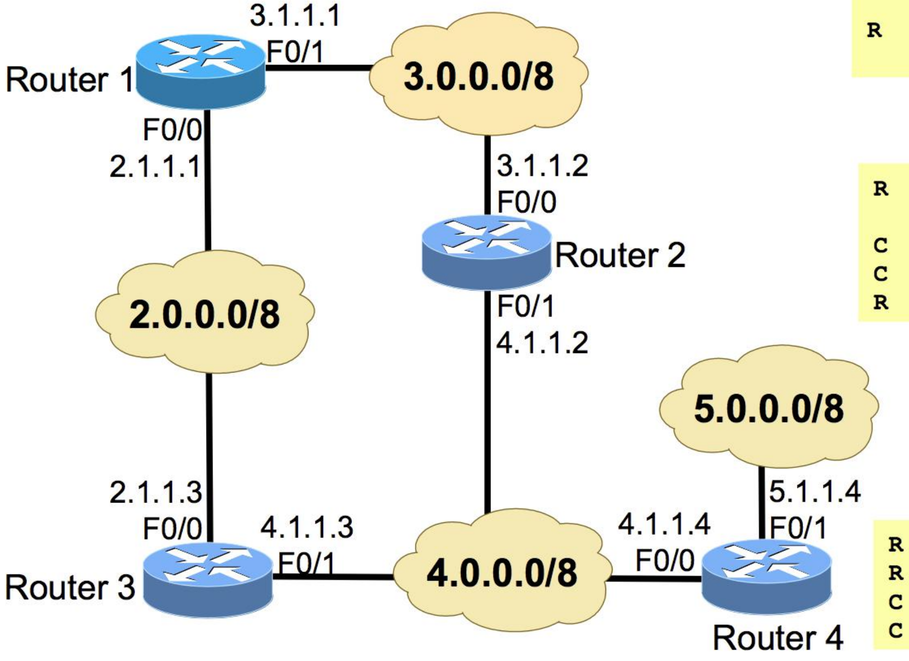
- Routing Information Protocol first version (RIPv1) RIPv1 is a dynamic, distance vector routing protocol.
- RIPv1 is a classful routing protocol that does not send the subnet mask in its updates.
  - RIPv1 does not support variable length network masks or discontinuous subnets.
  - RIPv1 automatically summarizes at the network boundary and cannot be configured not to.
- The cost of a path between a router and a network is given by the number of intermediary routers (hop count).
  - The maximum allowable hop count is 15 (16=infinite=unavailable).
- RIPv1 uses a 255.255.255.255 broadcast address,
  - All devices, including PCs and servers, must process the update packet.
- It has no authentication support.

Distance Vector

- Sem split horizon: cada router anuncia o vector distância completo por todas as interfaces.
- Com split horizon: em cada interface, o router anuncia apenas as redes destino par as quais essa interface não é usada no encaminhamento d pacotes de dados
- O split horizon diminui o tem de convergência das tabelas encaminhamento quando
- há alterações de topologia



Routing Tables with RIPv1 Entries




```
C     2.0.0.0/8 is directly connected, FastEthernet0/0
C     3.0.0.0/8 is directly connected, FastEthernet0/1
R     4.0.0.0/8 [120/1] via 3.1.1.2, 00:00:10, FastEthernet0/1
                [120/1] via 2.1.1.3, 00:00:00, FastEthernet0/0
R     5.0.0.0/8 [120/2] via 3.1.1.2, 00:00:08, FastEthernet0/1
                [120/2] via 2.1.1.3, 00:00:00, FastEthernet0/0
```

Router 1

```
R     2.0.0.0/8 [120/1] via 4.1.1.3, 00:00:15, FastEthernet0/1
                [120/1] via 3.1.1.1, 00:00:07, FastEthernet0/0
C     3.0.0.0/8 is directly connected, FastEthernet0/0
C     4.0.0.0/8 is directly connected, FastEthernet0/1
R     5.0.0.0/8 [120/1] via 4.1.1.4, 00:00:22, FastEthernet0/1
```

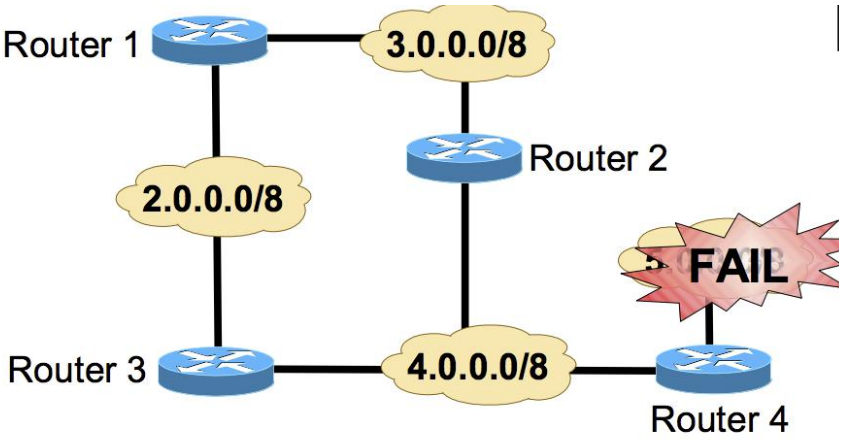
Router 2

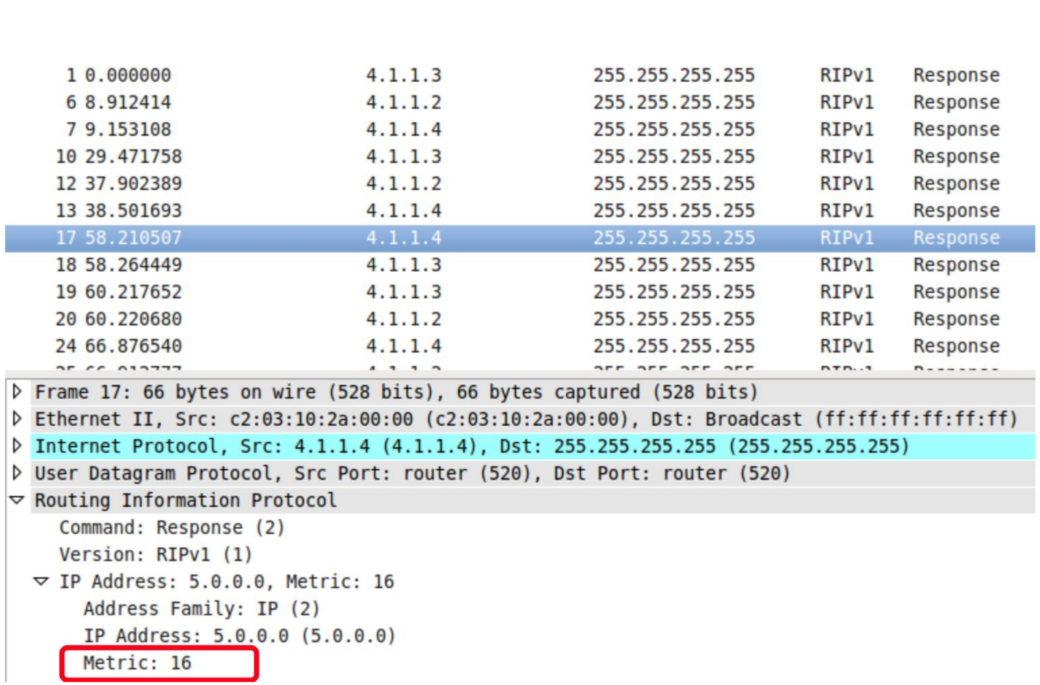
```
R     2.0.0.0/8 [120/1] via 4.1.1.3, 00:00:26, FastEthernet0/0
R     3.0.0.0/8 [120/1] via 4.1.1.2, 00:00:25, FastEthernet0/0
C     4.0.0.0/8 is directly connected, FastEthernet0/0
C     5.0.0.0/8 is directly connected, FastEthernet0/1
```

Router 4

```
C     2.0.0.0/8 is directly connected, FastEthernet0/0
R     3.0.0.0/8 [120/1] via 4.1.1.2, 00:00:09, FastEthernet0/1
                [120/1] via 2.1.1.1, 00:00:08, FastEthernet0/0
C     4.0.0.0/8 is directly connected, FastEthernet0/1
R     5.0.0.0/8 [120/1] via 4.1.1.4, 00:00:20, FastEthernet0/1
```

RIP behavior Upon a Link Fail



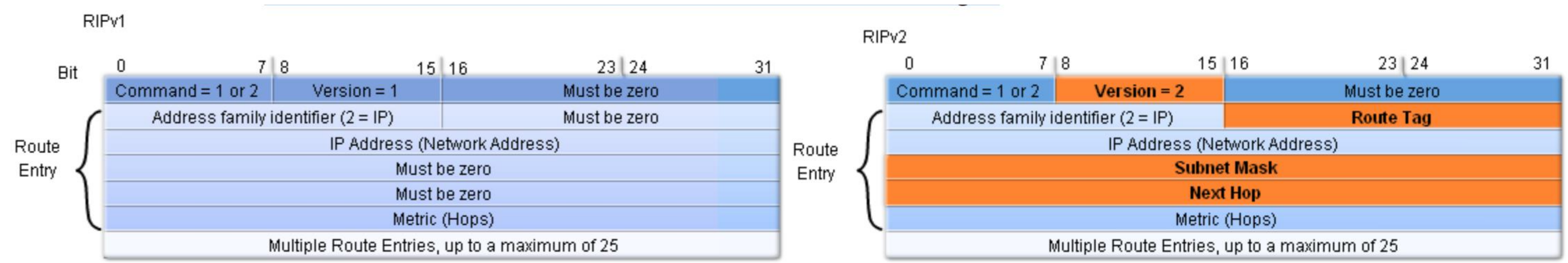


RIP Version 2

- RIPv2 is a classless distance vector routing protocol,
  - RIPv2 update packets include the network masks.
  - RIPv2 supports variable length network masks and discontinuous subnets.
  - RIPv2 automatically summarizes routes on classful network boundaries, however, it can be disabled.

- RIPv2 uses multicast addressing for more-efficient periodic updating on each interface,
  - Uses the 224.0.0.9 multicast address to advertise to other RIPv2 routers.
- RIPv2 also supports security between RIP routers using message-digest or clear-text authentication.

RIPv1 vs. RIPv2 Messages



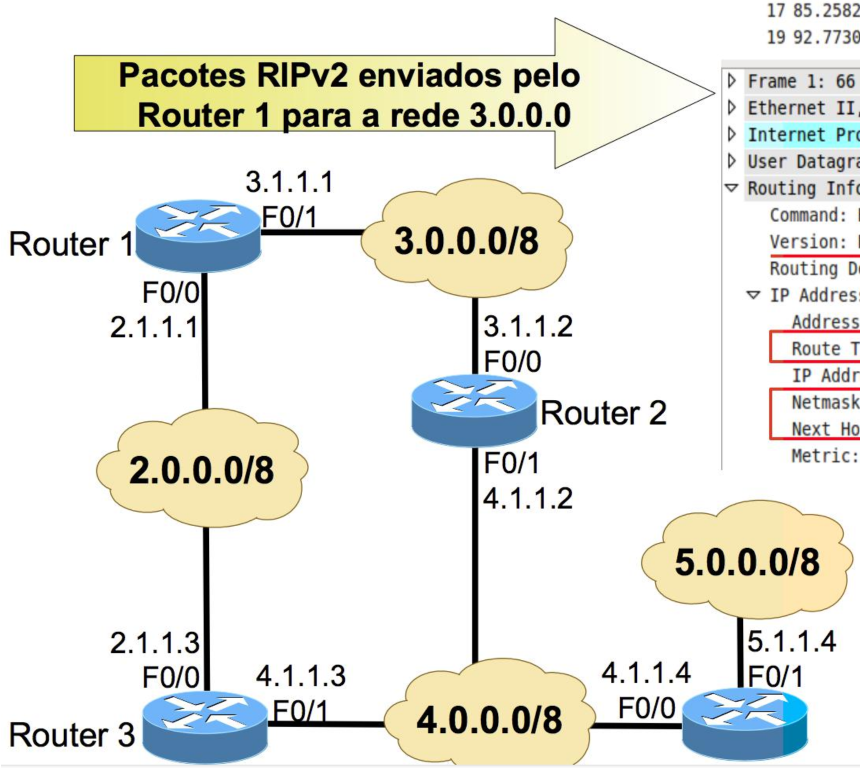
- New fields in RIPv2 messages:
  - Subnet mask
    - To make RIPv2 a classless routing protocol.
    - Supporting variable length network masks and discontinuous subnets.
  - Route tag
    - Attribute assigned to a route which must be preserved and re-advertised with a route.
    - Provide a method of separating routes for networks within the RIP routing domain from routes which may have been imported from other protocols (redistribution).
  - Next hop
    - IP address to where packets for this entry should be forwarded to.
    - Specifying 0.0.0.0 in this field indicates that packets for this entry should be forwarded to the router that sent this RIP message.

RIPv2 Response Messages

With *split horizon*

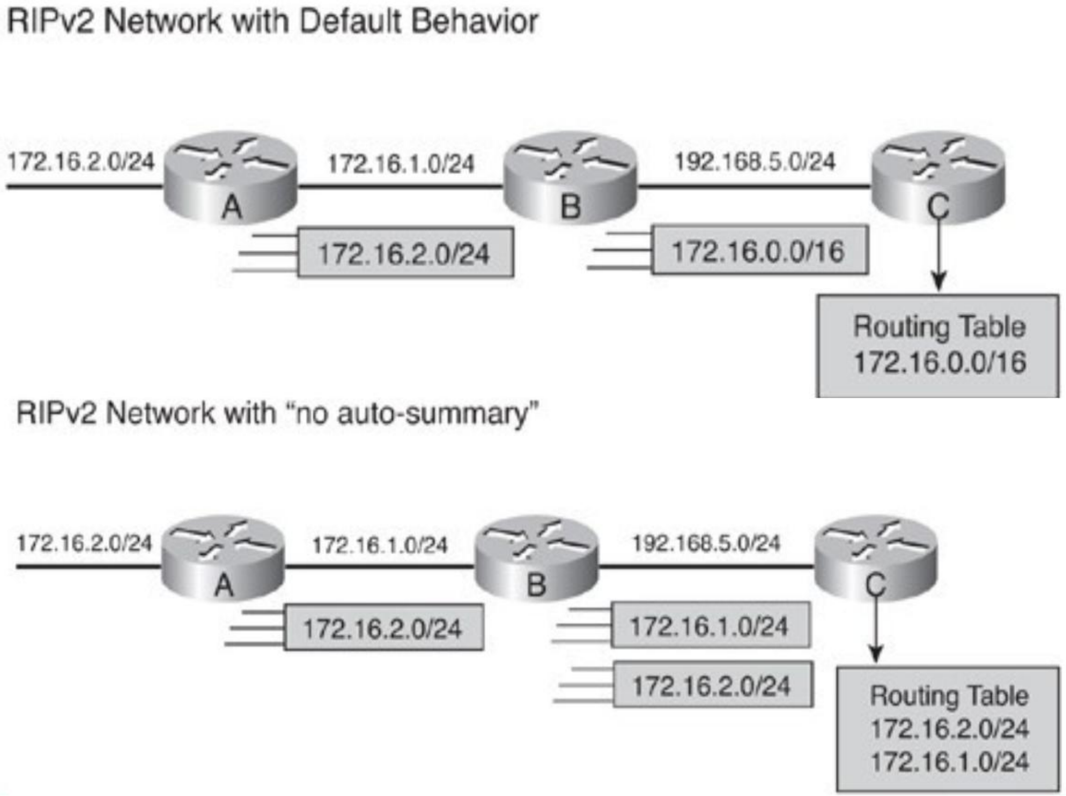
1 0.000000	3.1.1.1	224.0.0.9	RIPv2	Response
3 7.936722	3.1.1.2	224.0.0.9	RIPv2	Response
6 27.017157	3.1.1.1	224.0.0.9	RIPv2	Response
8 36.912644	3.1.1.2	224.0.0.9	RIPv2	Response
13 56.403552	3.1.1.1	224.0.0.9	RIPv2	Response
14 64.328612	3.1.1.2	224.0.0.9	RIPv2	Response
17 85.258230	3.1.1.1	224.0.0.9	RIPv2	Response
19 92.773090	3.1.1.2	224.0.0.9	RIPv2	Response

```
Frame 1: 66 bytes on wire (528 bits), 66 bytes captured (528 bits)
Ethernet II, Src: c2:00:10:2a:00:01 (c2:00:10:2a:00:01), Dst: IPv4mcast_00:00:09 (01:00:5e:00:00:09)
Internet Protocol, Src: 3.1.1.1 (3.1.1.1), Dst: 224.0.0.9 (224.0.0.9)
User Datagram Protocol, Src Port: router (520), Dst Port: router (520)
Routing Information Protocol
  Command: Response (2)
  Version: RIPv2 (2)
  Routing Domain: 0
  IP Address: 2.0.0.0, Metric: 1
    Address Family: IP (2)
    Route Tag: 0
    IP Address: 2.0.0.0 (2.0.0.0)
    Netmask: 255.0.0.0 (255.0.0.0)
    Next Hop: 0.0.0.0 (0.0.0.0)
    Metric: 1
```



RIPv2 Automatic Network-Boundary summarization

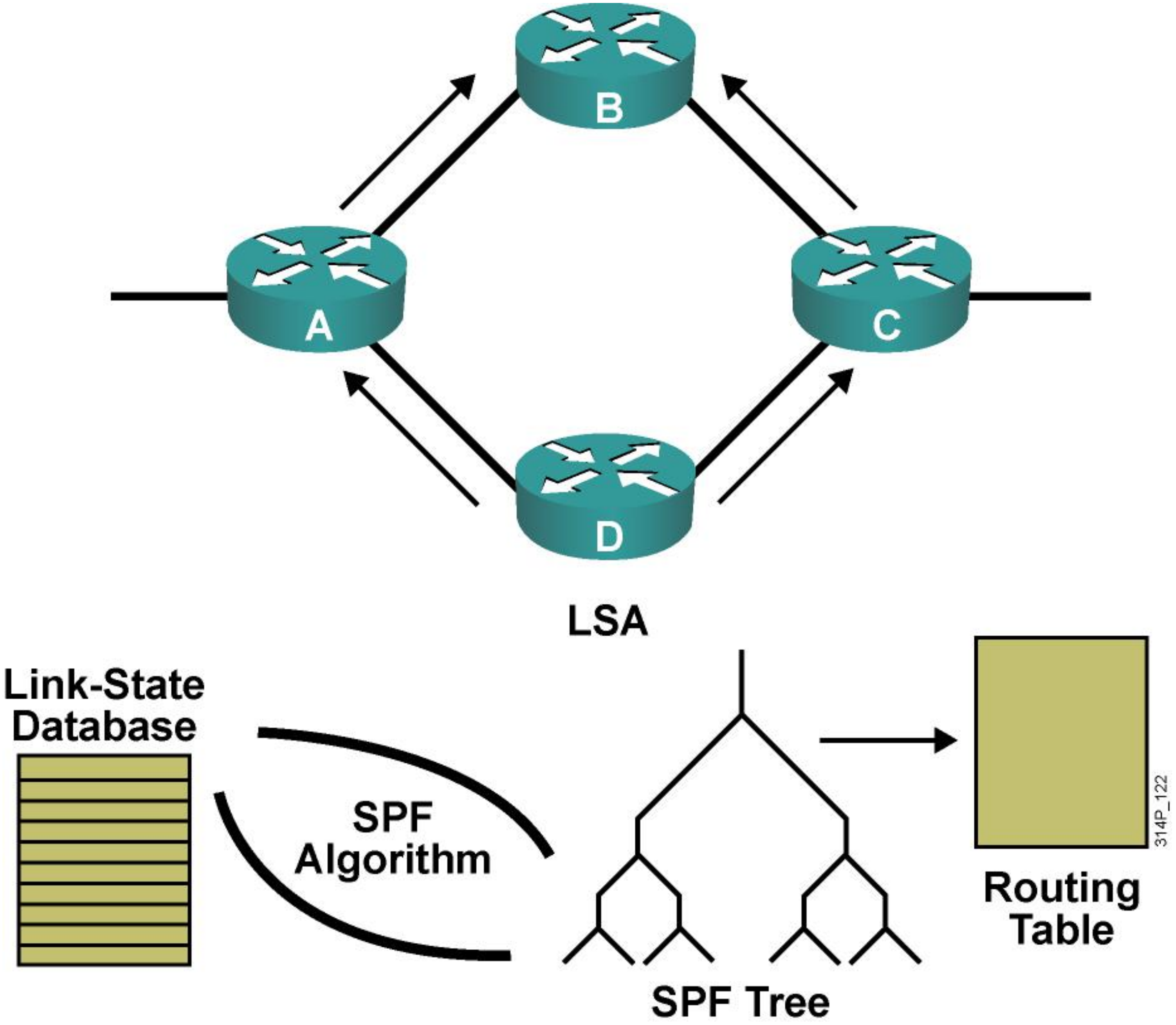
- By default RIPv2 automatic network summarization at classful boundaries, just like a classful protocol does.
- Automatic summarization lets RIPv2 be backward compatible with its predecessor (RIPv1).
- It can be turn off.



Configuring OSPF

Introducing the OSPF Protocol

Link-State Protocols



Link-State Data Structures

- Neighbor table:
  - Also known as the adjacency database
  - Contains list of recognized neighbors
- Topology table:
  - Typically referred to as LSDB
  - Contains all routers and their attached links in the area or network
  - Identical LSDB for all routers within an area
- Routing table:
  - Commonly named a forwarding database
  - Contains list of best paths to destinations

Link-State Routing Protocols

- Link-state routers recognize more information about the network than their distance vector counterparts.
- Each router has a full picture of the topology.
- Consequently, link-state routers tend to make more accurate decisions.

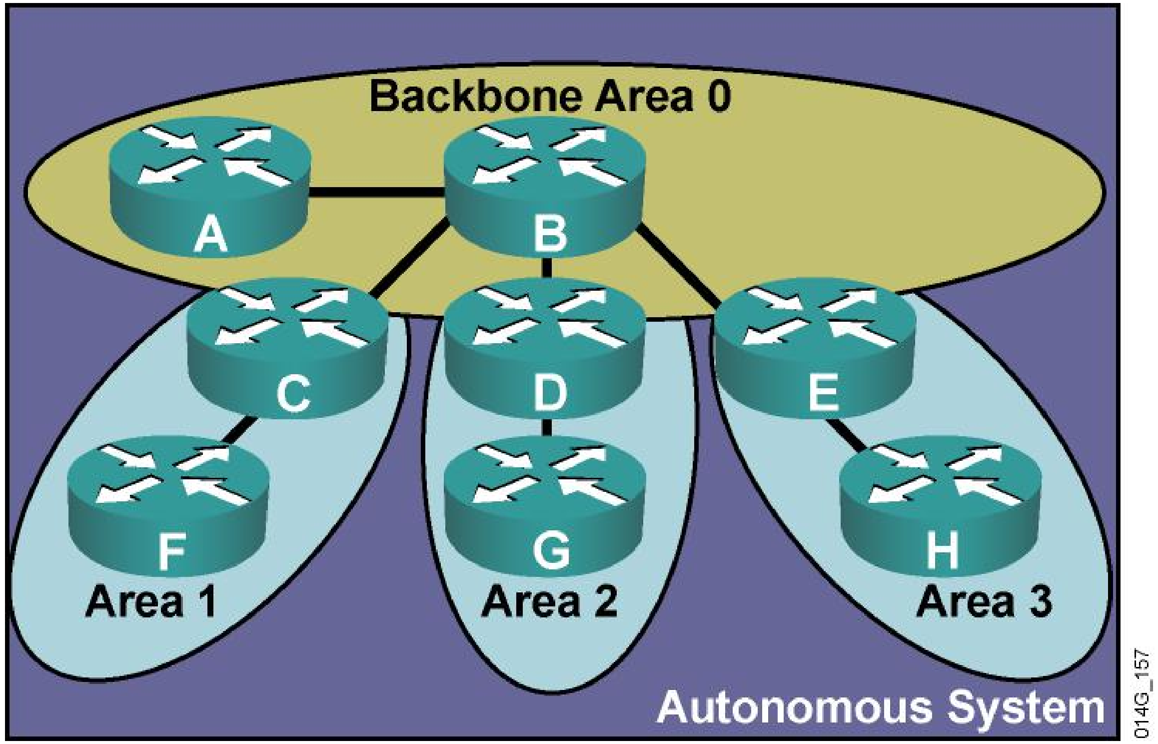
Link-State Data Structure: Network Hierarchy

- Link-state routing requires a hierachical network structure that is enforced by OSPF.
- This two-level hierarchy consists of the following:
  - Transit area (backbone or area 0)
  - Regular areas (nonbackbone areas)

OSPF Areas

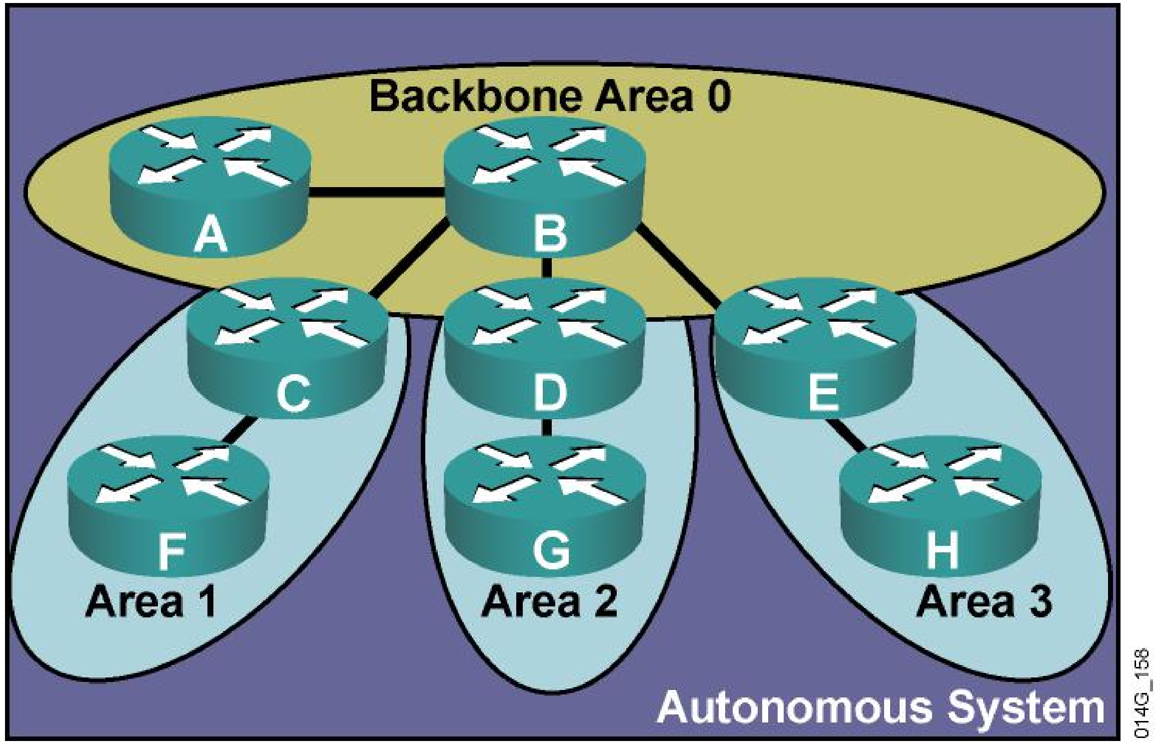
OSPF area characteristics:

- **Minimizes routing table entries**
- **Localizes impact of a topology change within an area**
- **Detailed LSA flooding stops at the area boundary**
- **Requires a hierarchical network design**



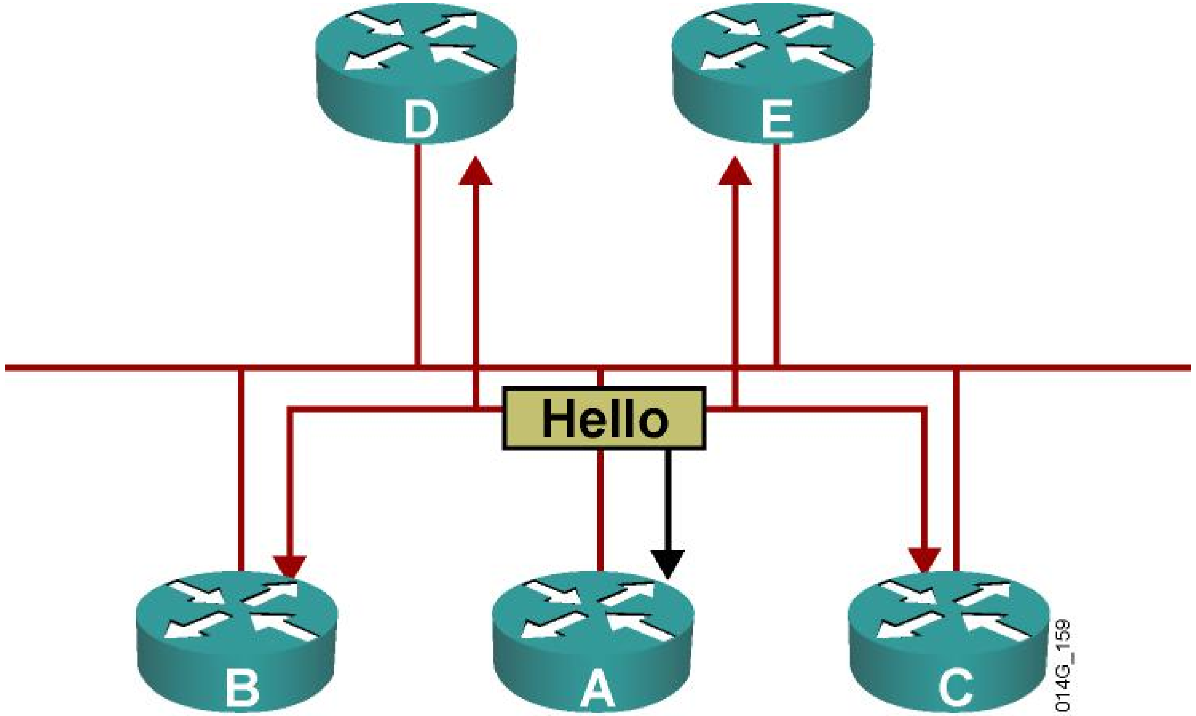
Area Terminology

- **Routers A and B are backbone routers.**
- **Backbone routers make up area 0.**
- **Routers C, D, and E are known as area border routers (ABRs).**
- **ABRs attach all other areas to area 0.**



OSPF Adjacencies

- Routers discover neighbors by exchanging hello packets.
- Routers declare neighbors to be up after checking certain parameters or options in the hello packet.



Forming OSPF Adjacencies

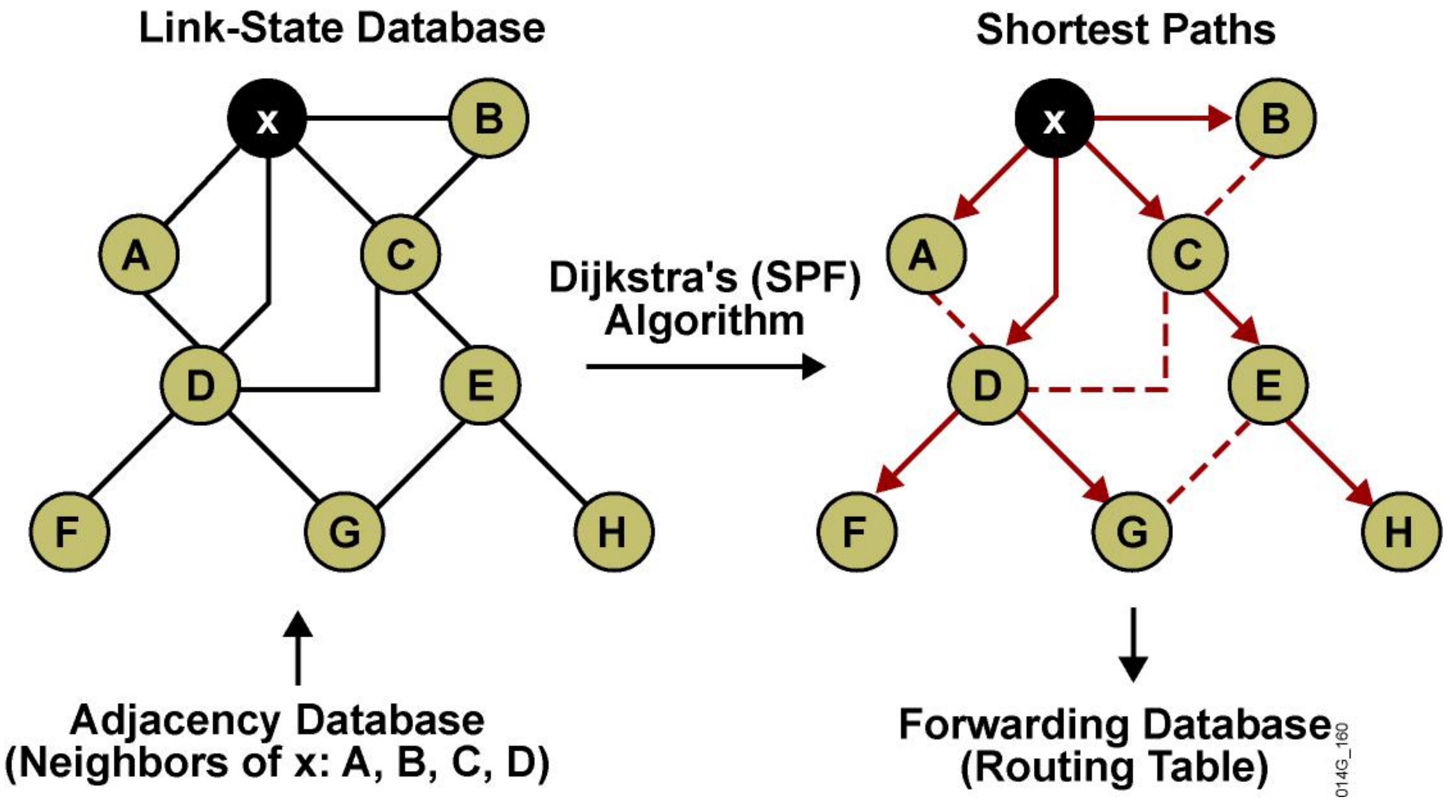
- Point-to-point WAN links:
  - Both neighbors become fully adjacent.
- LAN links:
  - Neighbors form a full adjacency with the DR and BDR.
  - Routers maintain two-way state with the other routers (DROTHERs).

- Routing updates and topology information are passed only between adjacent routers.
- Once an adjacency is formed, LSDBs are synchronized by exchanging LSAs.
- LSAs are flooded reliably throughout the area (or network).

OSPF Calculation

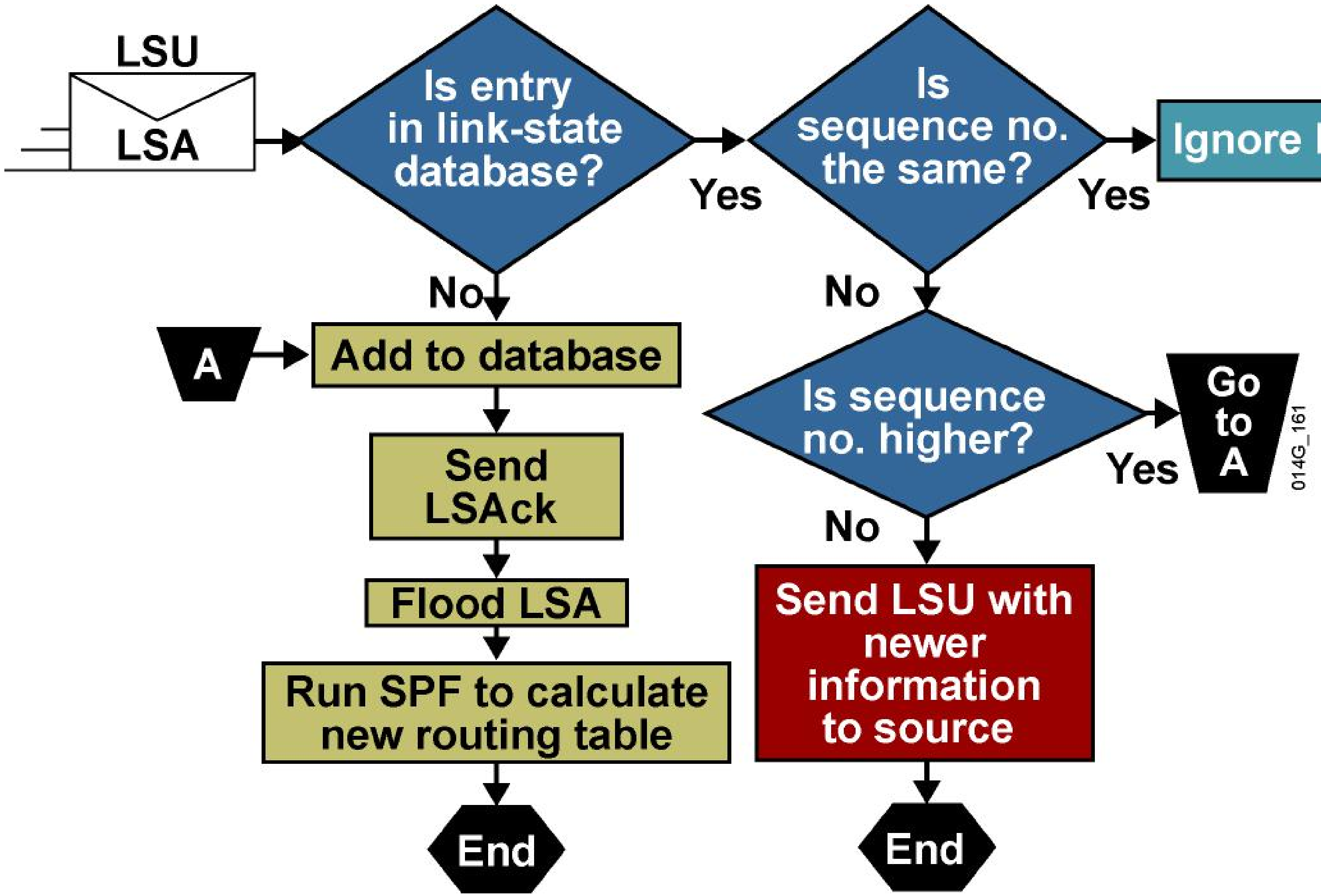
- Routers find the best paths to destinations by applying Dijkstra’s SPF algorithm to the link-state database as follows:
  - Every router in an area has the identical link-state database.
  - Each router in the area places itself into the root of the tree that is built.
  - The best path is calculated with respect to the lowest total cost of links to a specific destination.
  - Best routes are put into the forwarding database (routing table).

SPF Calculation



Assume all links are Ethernet, with an OSPF cost of 10.

Link-State Data Structures: LSA Operation



Summary

- Link-state routing protocols respond quickly to changes, send triggered updates when changes occur, and send periodic updates every 30 minutes.
- A two-tier hierarchical network structure is used by OSPF in which the network is divided into areas. This area structure is used to separate the LSDB into more manageable sizes.
- Adjacencies are built by OSPF routers using the Hello protocol. Over these logical adjacencies, LSUs are sent to exchange database information between adjacent OSPF routers.

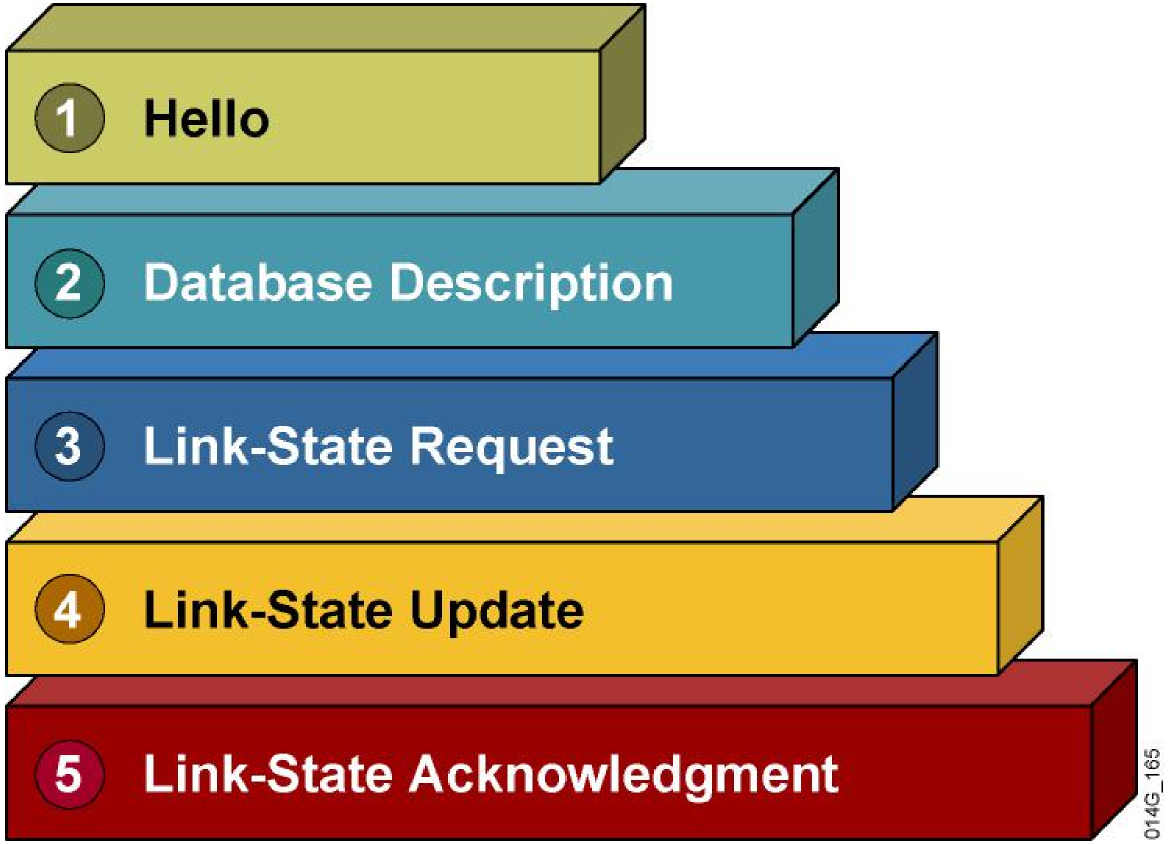
Summary (Cont.)

- Dijkstra’s SPF algorithm is used to calculate best paths for all destinations. SPF is run against the LSDB, and the outcome is a table of best paths, known as the routing table.
- After an LSA entry ages, the router that originated the entry sends an LSU about the network to verify that the link is still active. The LSU can contain one or more LSAs.

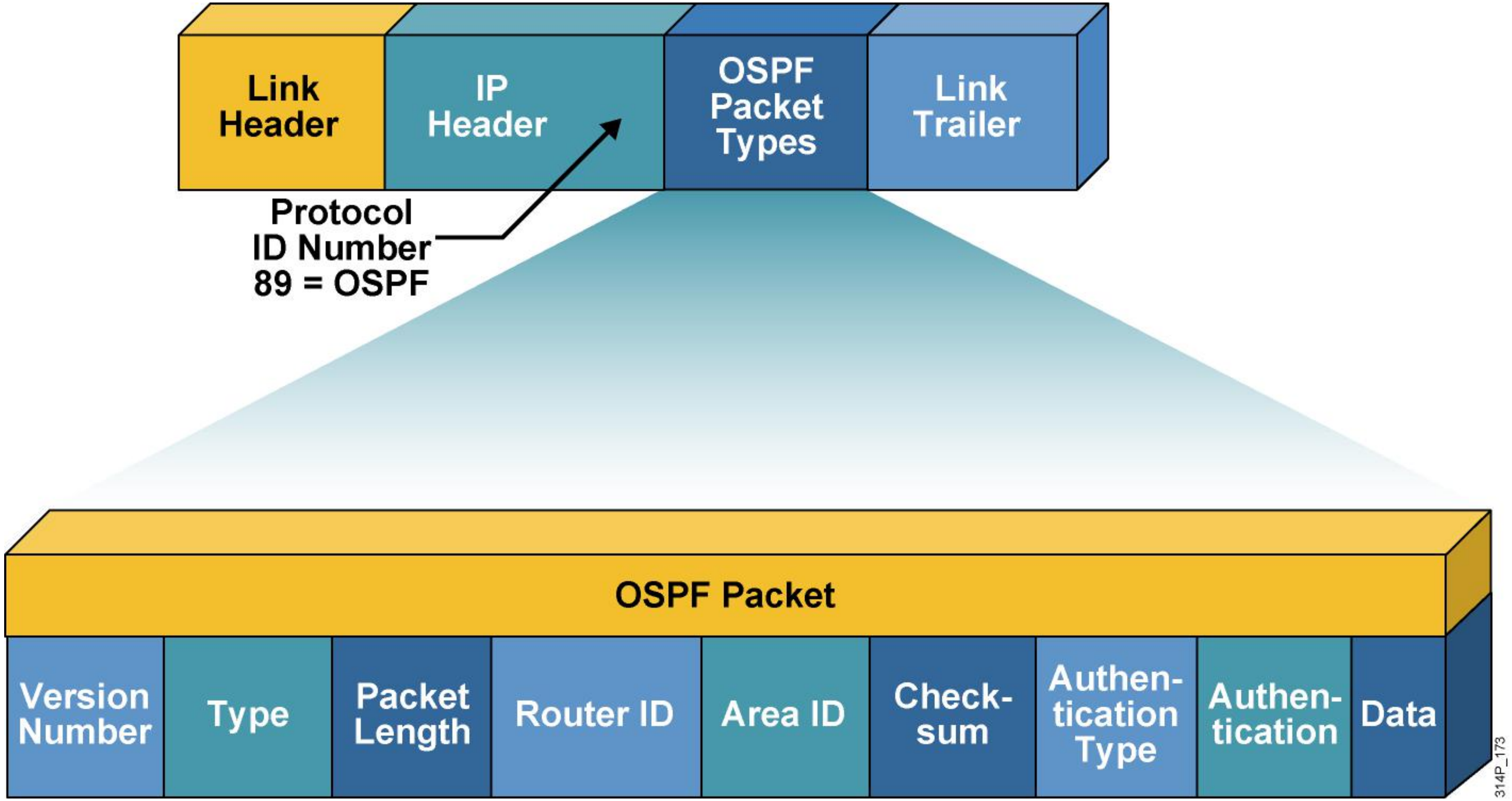
Configuring OSPF

OSPF Packet Types

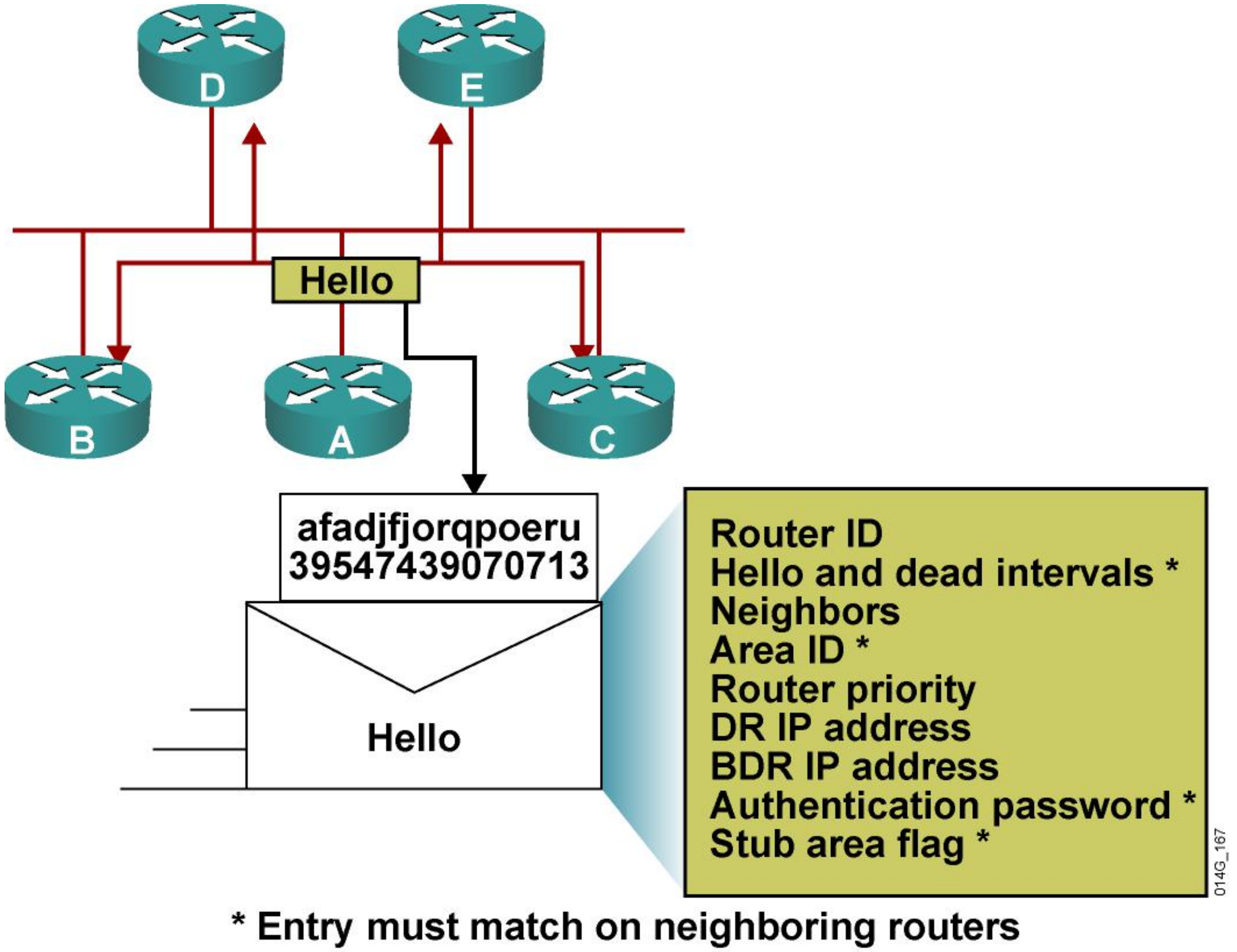
OSPF Packet Types



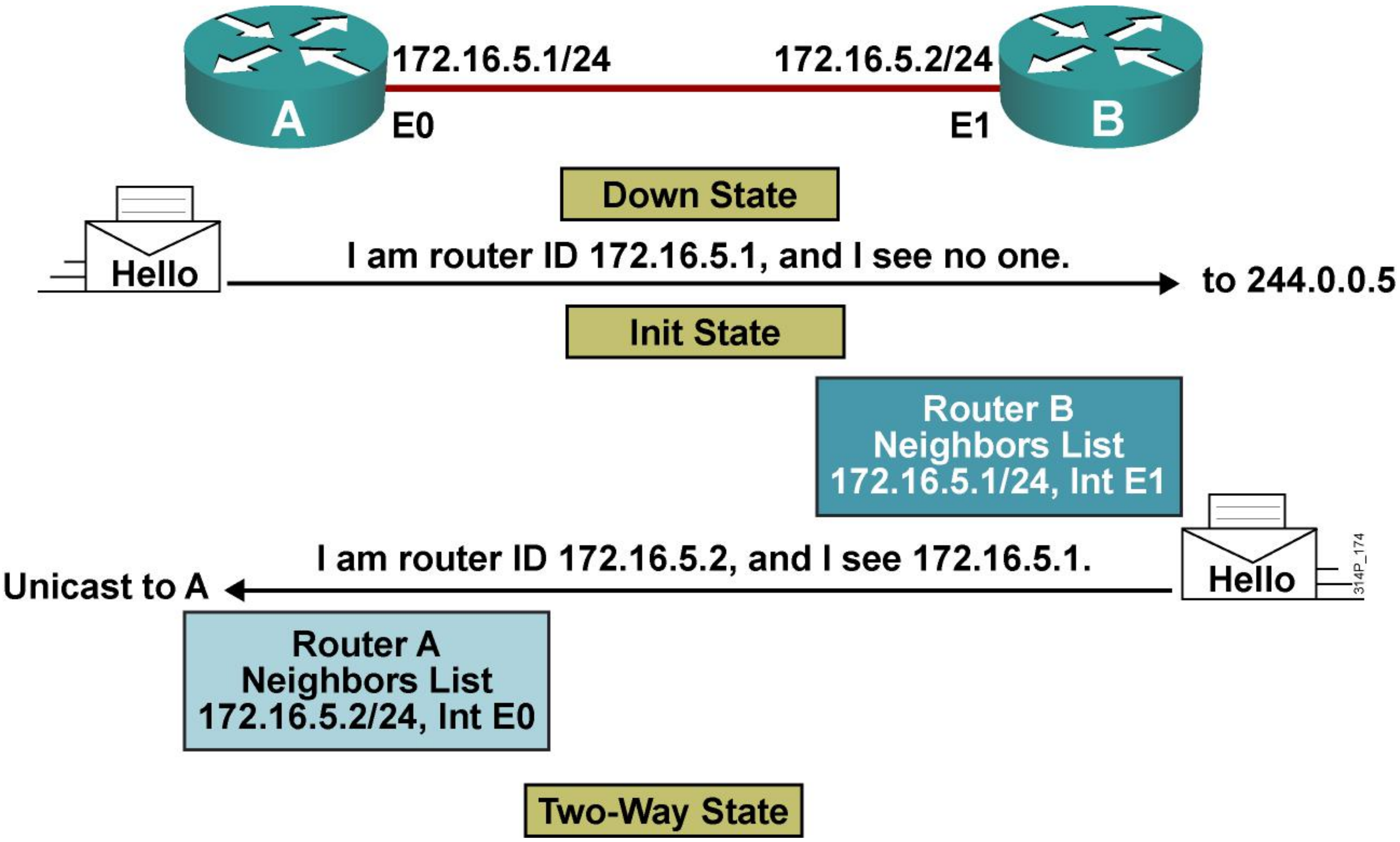
OSPF Packet Header Format



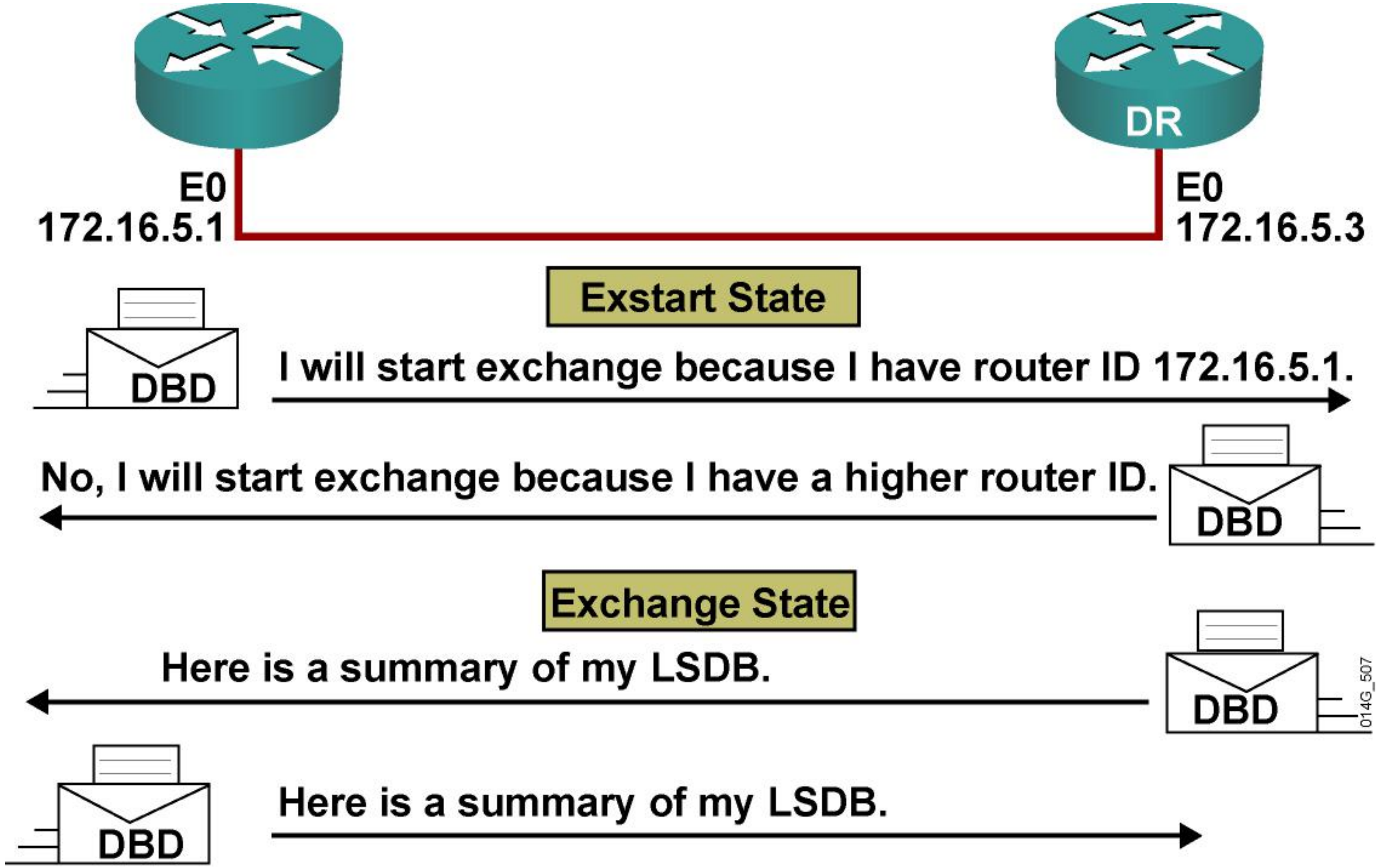
Neighborship: The Hello Packet



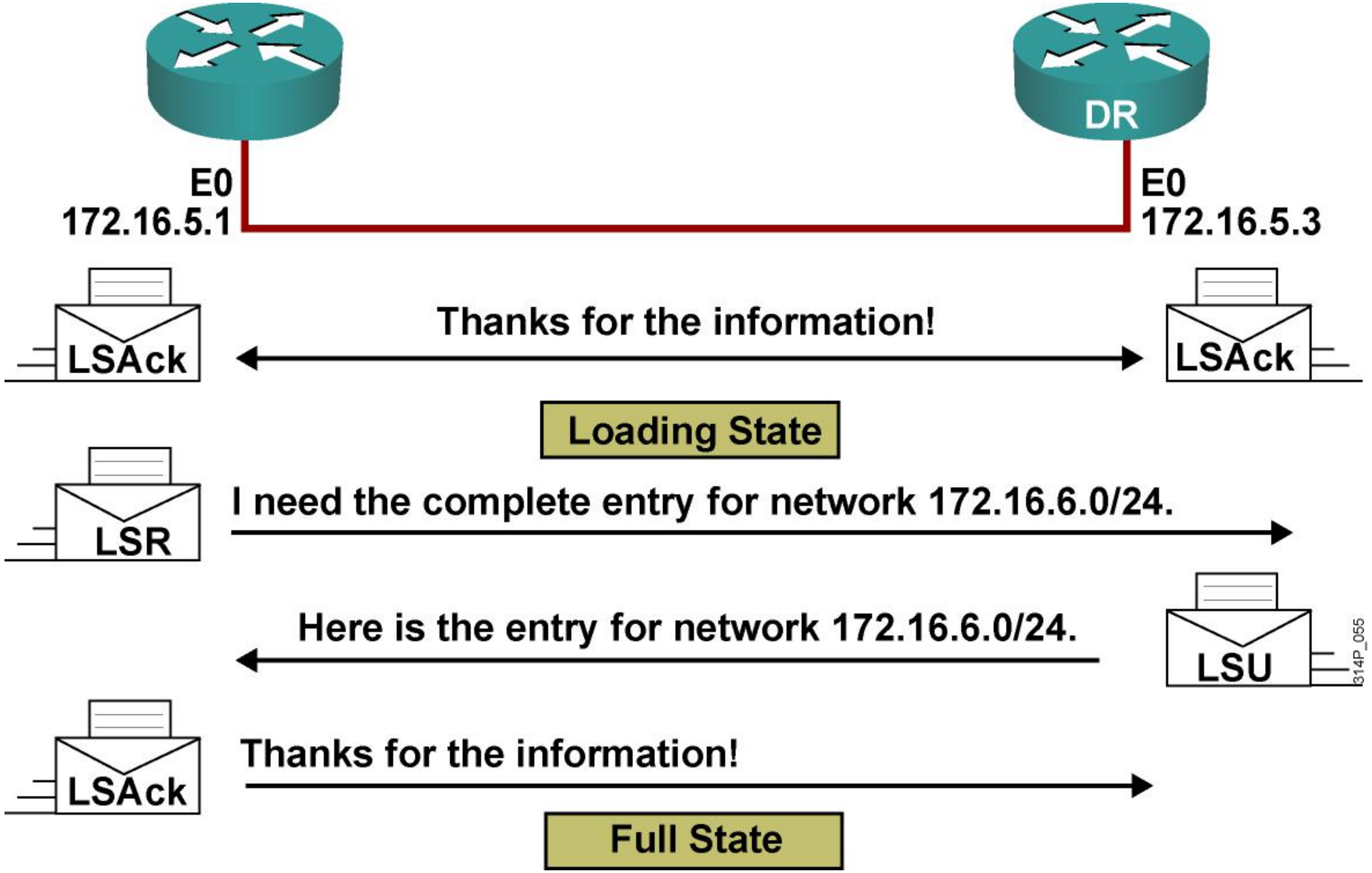
Establishing Bidirectional Communication



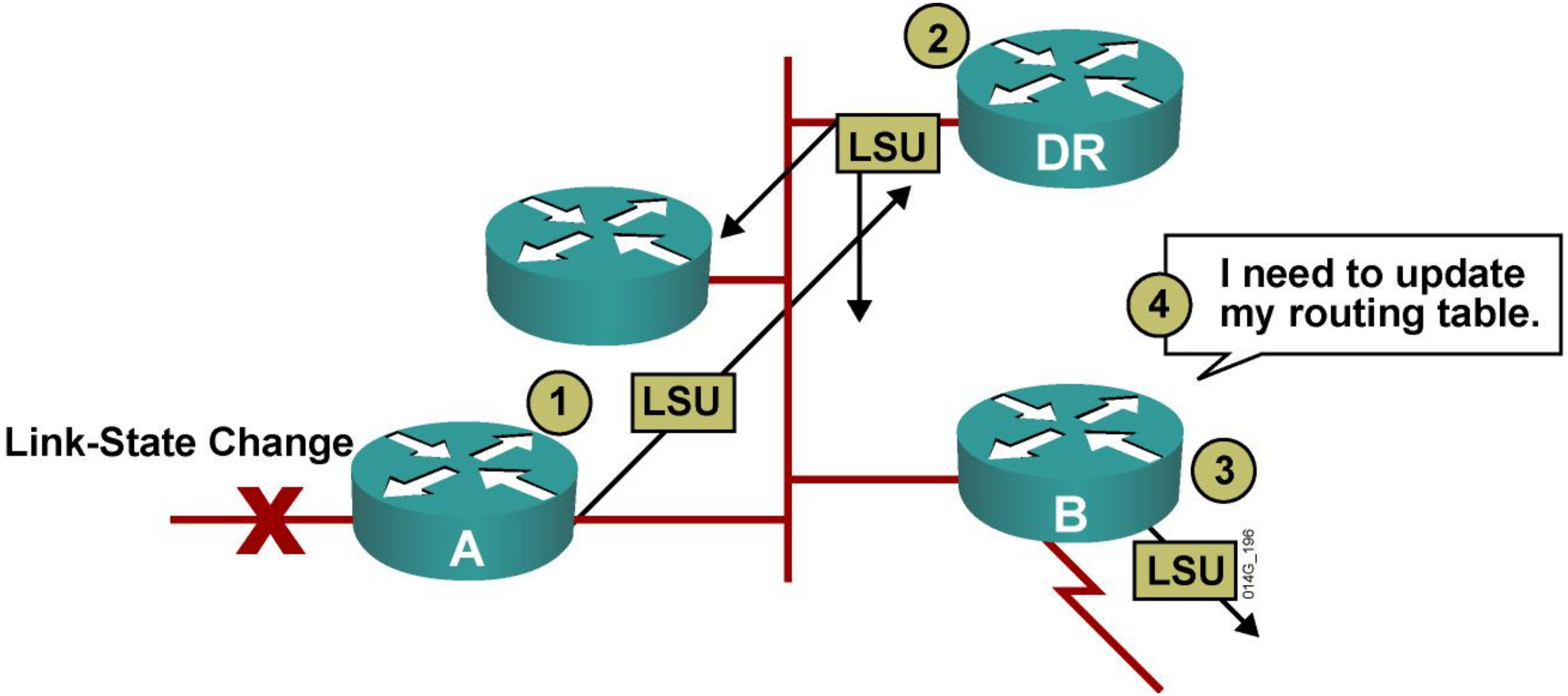
Discovering the Network Routes



Adding the Link-State Entries



Maintaining Routing Information



- **Router A notifies all OSPF DRs on 224.0.0.6.**
- **DR notifies others on 224.0.0.5.**

LSA Sequence Numbering

- Each LSA in the LSDB maintains a sequence number.
- The sequence numbering scheme is a 4-byte number that begins with 0x80000001 and ends with 0x7FFFFFFF.
- OSPF floods each LSA every 30 minutes to maintain proper database synchronization. Each time the LSA is flooded, the sequence number is incremented by one.
- Ultimately, an LSA sequence number will wrap around to 0x80000001. When this occurs, the existing LSA is prematurely aged to the maximum age (one hour) and flushed.
- When a router encounters two instances of an LSA, it must determine which is more recent. The LSA having the newer (higher) LS a sequence number is more recent.

LSA Sequence Numbers and Maximum Age

```
RTC# show ip ospf database

             OSPF Router with ID (192.168.1.67) (Process ID 10)
                      Router Link States (Area 1)
Link ID             ADV Router  Age  Seq#           Checksum  Link count
192.168.1.67        192.168.1.67        48   0x80000008  0xB112    2
192.168.2.130       192.168.2.130       212  0x80000006  0x3F44    2
<output omitted>
```

- Every OSPF router announces a router LSA for those interfaces that it owns in that area.
- Router with link ID 192.168.1.67 has been updated eight times; the last update was 48 seconds ago.

debug ip ospf packet

Debug of a single packet

```
R1#debug ip ospf packet
OSPF packet debugging is on
R1#
*Feb 16 11:03:51.206: OSPF: rcv. v:2 t:1 l:48 rid:10.0.0.12
      aid:0.0.0.1 chk:D882 aut:0 auk: from Serial0/0/0.2
```

- **Shows fields in OSPF header**

Summary

- There are five OSPF packet types: hello, DBD, LSU, LSR, and LSAck.
- The Hello protocol forms logical neighbor adjacency relationships. A DR may be required to coordinate adjacency formations.
- The exchange protocol passes through several states (down, init, two-way, exstart, and exchange) before finally reaching the goal of full state. Full state means that databases are synchronized with adjacent routers.

Summary (Cont.)

- LSAs are sent on change but are also sent every 30 minutes to ensure database integrity. The maximum time that an LSA will stay in the database, without an update, is 1 hour. The LSA sequence number is incremented every time it is advertised.

- Each LSA in the LSDB has a sequence number, which is incremented by one each time the LSA is flooded. When a router encounters two instances of an LSA, it must determine which is more recent. The LSA having the newer (higher) LSA sequence number is more recent.
- Use the debug ip ospf packet command to verify that OSPF packets are flowing properly between two routers.

Configuring OSPF

Configuring OSPF Routing

Configuring Basic OSPF

Router(config)#

router ospf *process-id* **[vrf** *vpn-name***]**

- **Enables one or more OSPF routing processes**

Router(config-router)#

network *ip-address wildcard-mask* **area** *area-id*

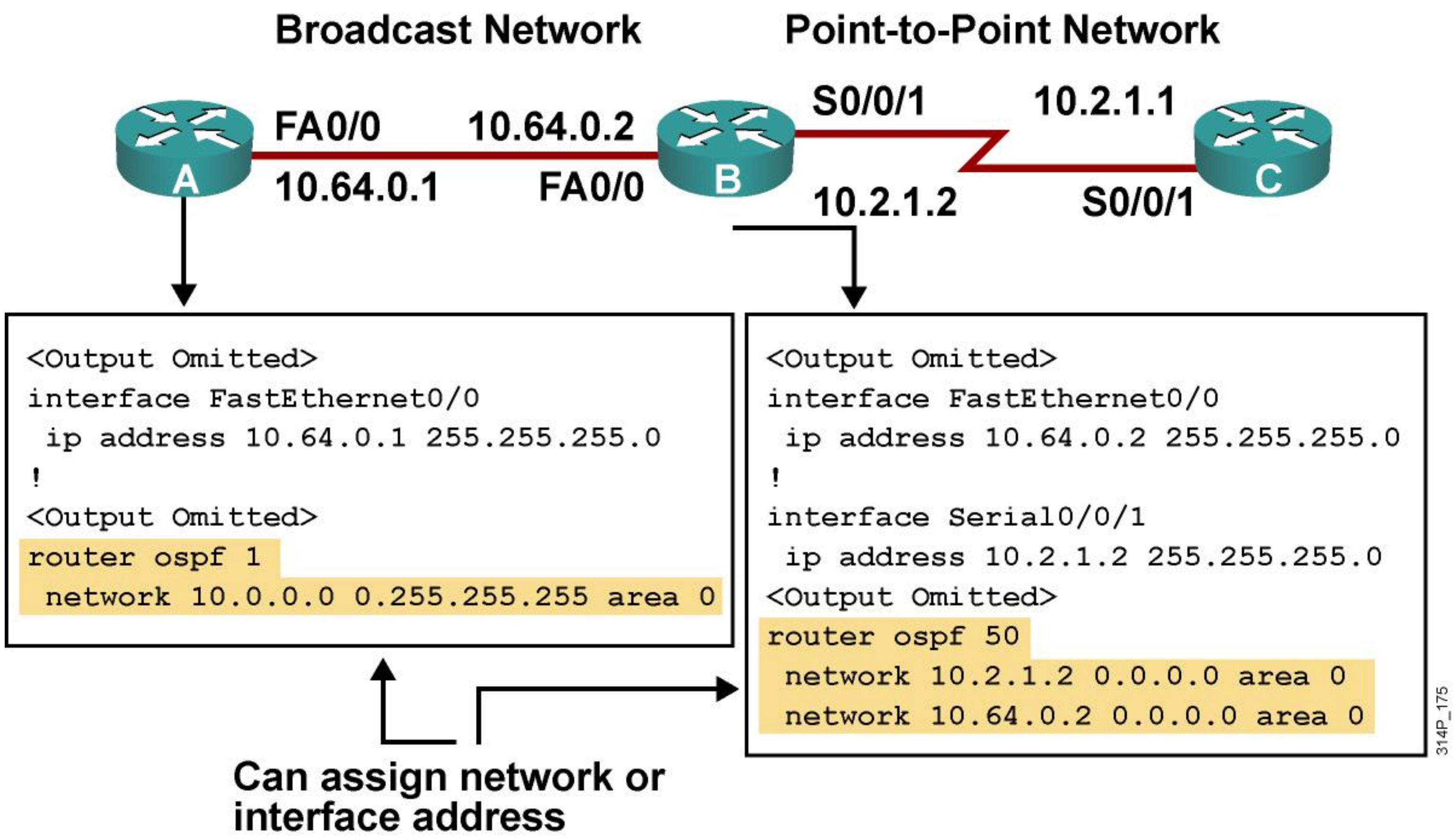
- **Defines the interfaces that OSPF will run on**

Router(config-if)#

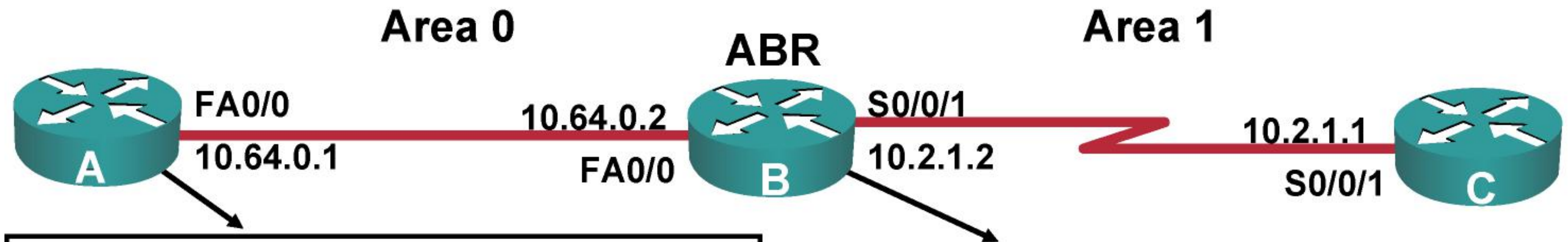
ip ospf *process-id* **area** *area-id* **[secondaries none]**

- **Optional method to enable OSPF explicitly on an interface**

Configuring OSPF on Internal Routers of a Single Area



Configuring OSPF for Multiple Areas



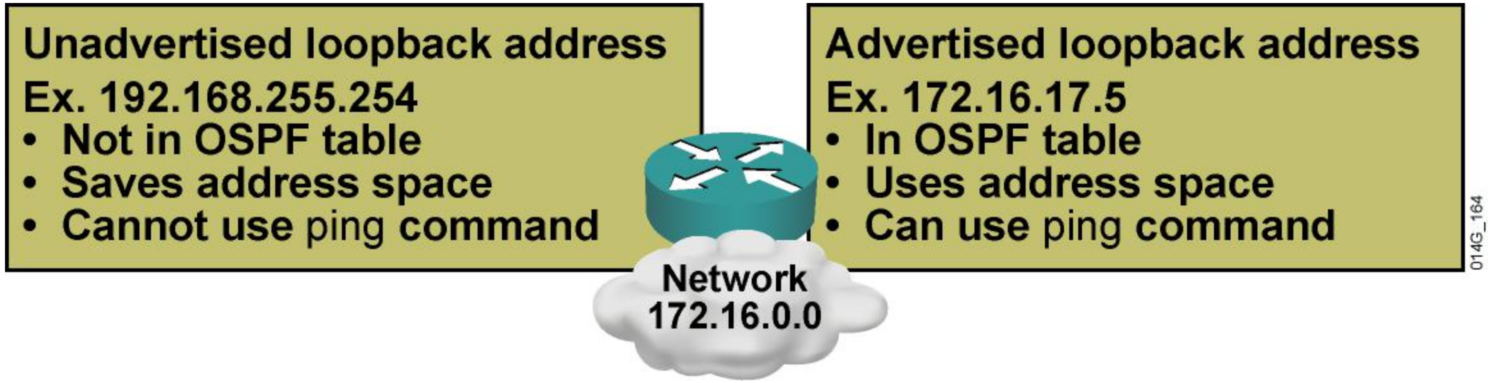

```
<output omitted>
interface FastEthernet0/0
 ip address 10.64.0.1 255.255.255.0

<output omitted>
router ospf 1
 network 10.0.0.0 0.255.255.255 area 0
```

```
<output omitted>
interface FastEthernet0/0
 ip address 10.64.0.2 255.255.255.0

interface Serial0/0/1
 ip address 10.2.1.2 255.255.255.0
 ip ospf 50 area 1

<output omitted>
router ospf 50
 network 10.64.0.2 0.0.0.0 area 0
```

OSPF Router ID

- The router is known to OSPF by the OSPF router ID number.
- LSDBs use the OSPF router ID to differentiate one router from the next.
- By default, the router ID is the highest IP address on an active interface at the moment of OSPF process startup.
- A loopback interface can override the OSPF router ID. If a loopback interface exists, the router ID is the highest IP address on any active loopback interface.
- The OSPF router-id command can be used to override the OSPF router ID.
- Using a loopback interface or a router-id command is recommended for stability.

Loopback Interfaces



```
Router(config)#interface loopback 0
Router(config-if)#ip address 172.16.17.5 255.255.255.255
```

- **If the OSPF process is already running, the router must be reloaded or the OSPF process must be removed and reconfigured before the new loopback address will take effect.**

OSPF router-id Command

Router(config-router)#

router-id *ip-address*

- **This command is configured under the** router ospf [*process-id*] **command.**
- **Any unique arbitrary 32-bit value in an IP address format (dotted decimal) can be used.**
- **If this command is used on an OSPF process that is already active, then the new router ID is used after the next reload or manual OSPF process restart using:**

Router#

clear ip ospf process

```
Router(config)#router ospf 1
Router(config-router)#router-id 172.16.1.1

Router#clear ip ospf process
```

OSPF Router ID Verification

```
RouterB#sh ip ospf
 Routing Process "ospf 50" with ID 10.64.0.2
<output omitted>

 Number of areas in this router is 2. 2 normal 0 stub 0 nssa
 Number of areas transit capable is 0
 External flood list length 0
    Area BACKBONE(0)
    Area BACKBONE(0)
        Area has no authentication
        SPF algorithm last executed 00:01:25.028 ago
        SPF algorithm executed 7 times
<output omitted>

  Area 1
        Number of interfaces in this area is 1
        Area has no authentication
        SPF algorithm last executed 00:00:54.636 ago
        SPF algorithm executed 3 times
<output omitted>
```

Verifying OSPF Operation

Router#

show ip protocols

- **Verifies the configured IP routing protocol processes, parameters, and statistics**

Router#

show ip route ospf [*process-id* **]**

- **Displays all OSPF routes learned by the router**

Router#

show ip ospf interface [*type number***]**

- **Displays the OSPF router ID, area ID, and adjacency information**

Verifying OSPF Operation (Cont.)

Router#

show ip ospf

- **Displays the OSPF router ID, timers, and statistics**

Router#

show ip ospf neighbor [*type number***] [***neighbor-id***] [detail]**

- **Displays information about the OSPF neighbors, including DR and BDR information on broadcast networks**

Example: The show ip route ospf Command

```
RouterA#show ip route ospf
     10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
O IA    10.2.1.0/24 [110/782] via 10.64.0.2, 00:03:05, FastEthernet0/0
RouterA#
```

Example: The show ip ospf interface Command

```
RouterA#show ip ospf interface fastEthernet 0/0
FastEthernet0/0 is up, line protocol is up
  Internet Address 10.64.0.1/24, Area 0
  Process ID 1, Router ID 10.64.0.1, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DROTHER, Priority 0
  Designated Router (ID) 10.64.0.2, Interface address 10.64.0.2
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    oob-resync timeout 40
    Hello due in 00:00:04
  Supports Link-local Signaling (LLS)
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 4
  Last flood scan time is 0 msec, maximum is 4 msec
  Neighbor Count is 1, Adjacent neighbor count is 1
    Adjacent with neighbor 10.64.0.2  (Designated Router)
  Suppress hello for 0 neighbor(s)
```

Example: The show ip ospf neighbor Command

```
RouterB# show ip ospf neighbor

Neighbor ID   Pri   State          Dead Time   Address     Interface
10.64.0.1     0     FULL/DROTHER   00:00:30    10.64.0.1  FastEthernet0/0
10.2.1.1      0     FULL/  -       00:00:34    10.2.1.1   Serial0/0/1
```

```
RouterB# show ip ospf neighbor detail
 Neighbor 10.64.0.1, interface address 10.64.0.1
    In the area 0 via interface FastEthernet0/0
    Neighbor priority is 0, State is FULL, 16 state changes
    DR is 10.64.0.2 BDR is 0.0.0.0
<output omitted>

Neighbor 10.2.1.1, interface address 10.2.1.1
    In the area 1 via interface Serial0/0/1
    Neighbor priority is 0, State is FULL, 6 state changes
    DR is 0.0.0.0 BDR is 0.0.0.0
<output omitted>
```

Summary

- Configuration of OSPF is a two-step process:
  - Enter OSPF configuration with the router ospf command.
  - Use the network command to describe which interfaces will run OSPF in which area.
- OSPF selects a router ID at startup time:
  - The router ID's specified in the router-id command under the OSPF process.
  - Otherwise, the highest IP address of a loopback interface, if there are any, is used.
  - By default, the highest IP address of all active interfaces
- Use the show ip ospf command to verify the router ID.
- Use the show ip protocols, show ip route ospf, show ip ospf interface, show ip ospf, and show ip ospf neighbor commands to verify OSPF operation.

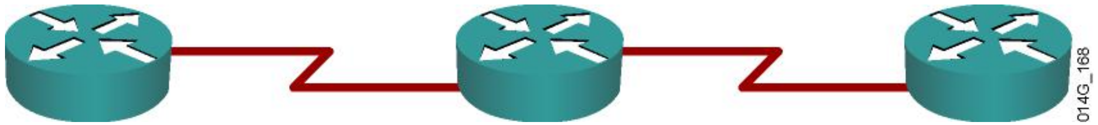
Configuring OSPF

OSPF Network Types

OSPF Network Types

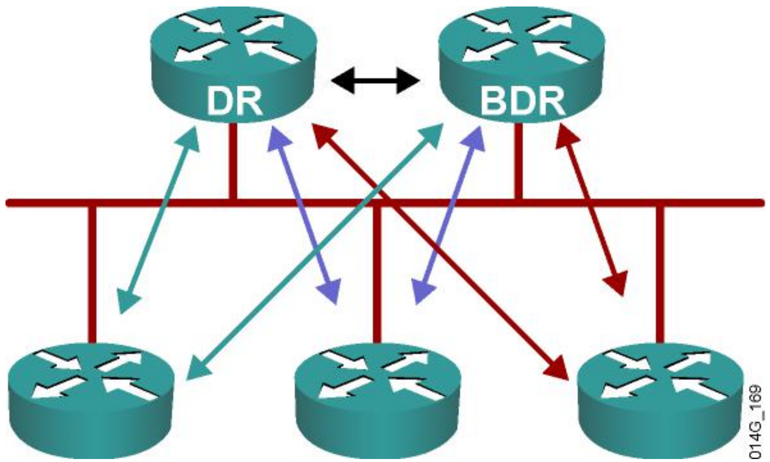
- The three types of networks defined by OSPF are:
  - Point-to-point: A network that joins a single pair of routers.
  - Broadcast: A multiaccess broadcast network, such as Ethernet.
  - Nonbroadcast multiaccess (also called NBMA): A network that interconnects more than two routers but that has no broadcast capability. Frame Relay, ATM, and X.25 are examples of NBMA networks.
    - Five modes of OSPF operation are available for NBMA networks.

Point-to-Point Links



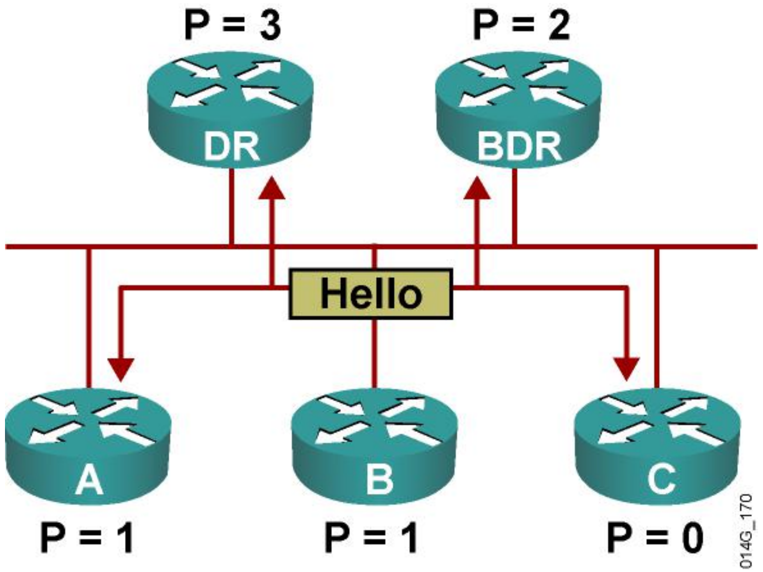
- Usually a serial interface running either PPP or HDLC.
- May also be a point-to-point subinterface running Frame Relay or ATM.
- No DR or BDR election required.
- OSPF autodetects this interface type.
- OSPF packets are sent using multicast 224.0.0.5.

Multiaccess Broadcast Network



- Generally these are, LAN technologies like Ethernet and Token Ring.
- DR and BDR selection are required.
- All neighbor routers form full adjacencies with the DR and BDR only.
- Packets to the DR and the BDR use 224.0.0.6.
- Packets from DR to all other routers use 224.0.0.5.

Electing the DR and BDR



- Hello packets are exchanged via IP multicast.
- The router with the highest OSPF priority is selected as the DR. The router with the second-highest priority value is the BDR.
- Use the OSPF router ID as the tiebreaker.
- The DR election is nonpreemptive.

Setting Priority for DR Election

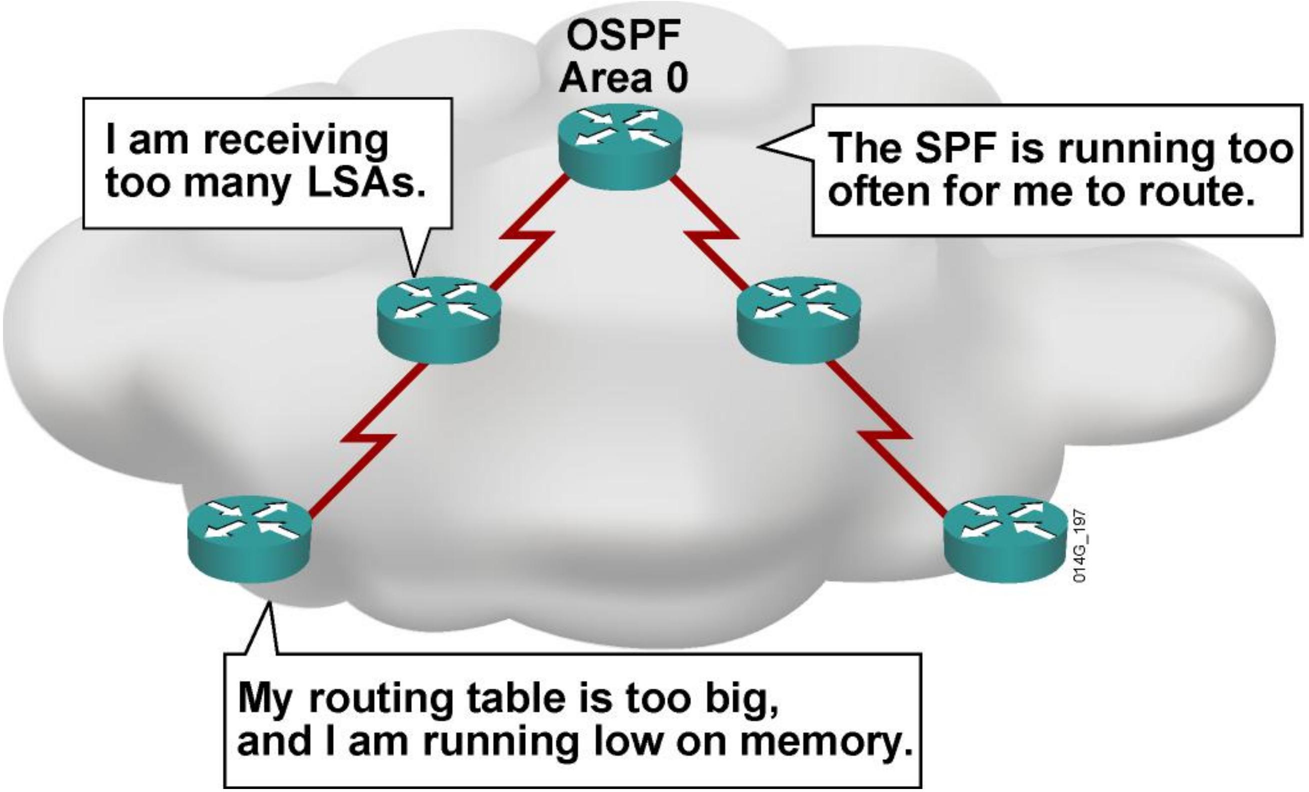
Router(config-if)#

ip ospf priority *number*

- This interface configuration command assigns the OSPF priority to an interface.
- Different interfaces on a router may be assigned different values.
- The default priority is 1. The range is from 0 to 255.
- 0 means the router cannot be the DR or BDR.
- A router that is not the DR or BDR is DROTHER.

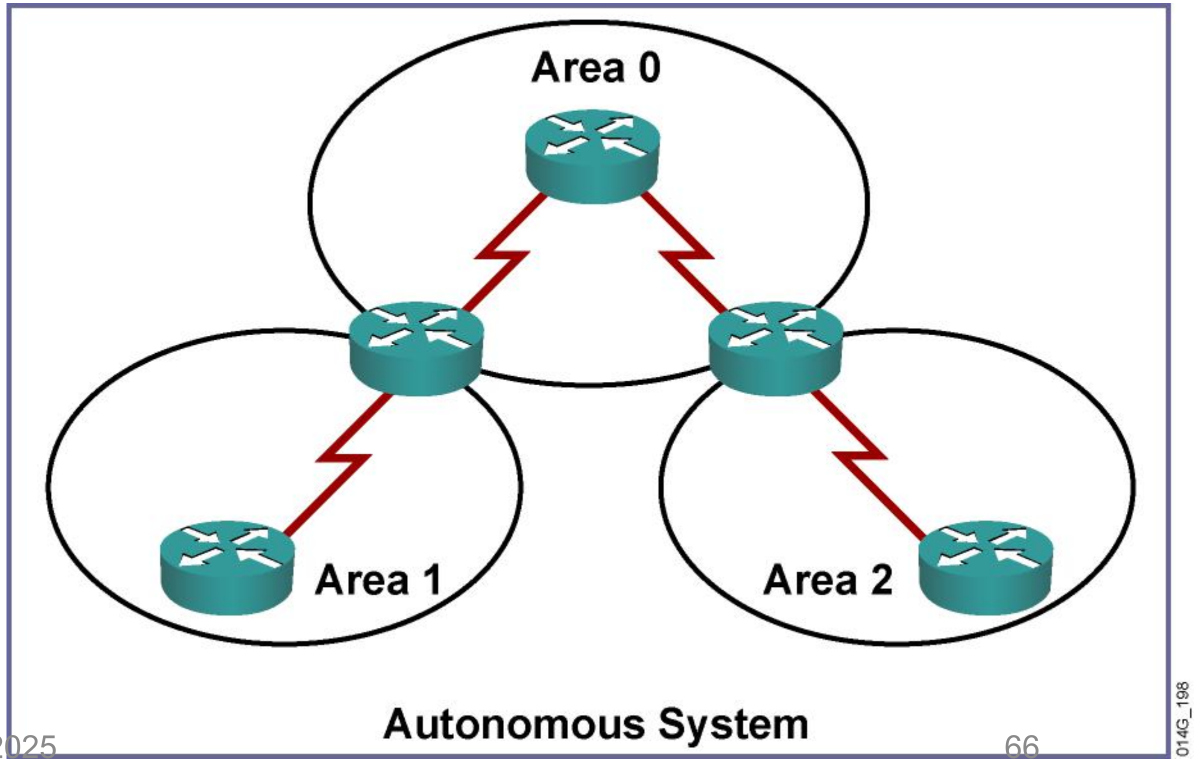
Configuring OSPF
Link-State Advertisements

Issues with Maintaining a Large OSPF Network

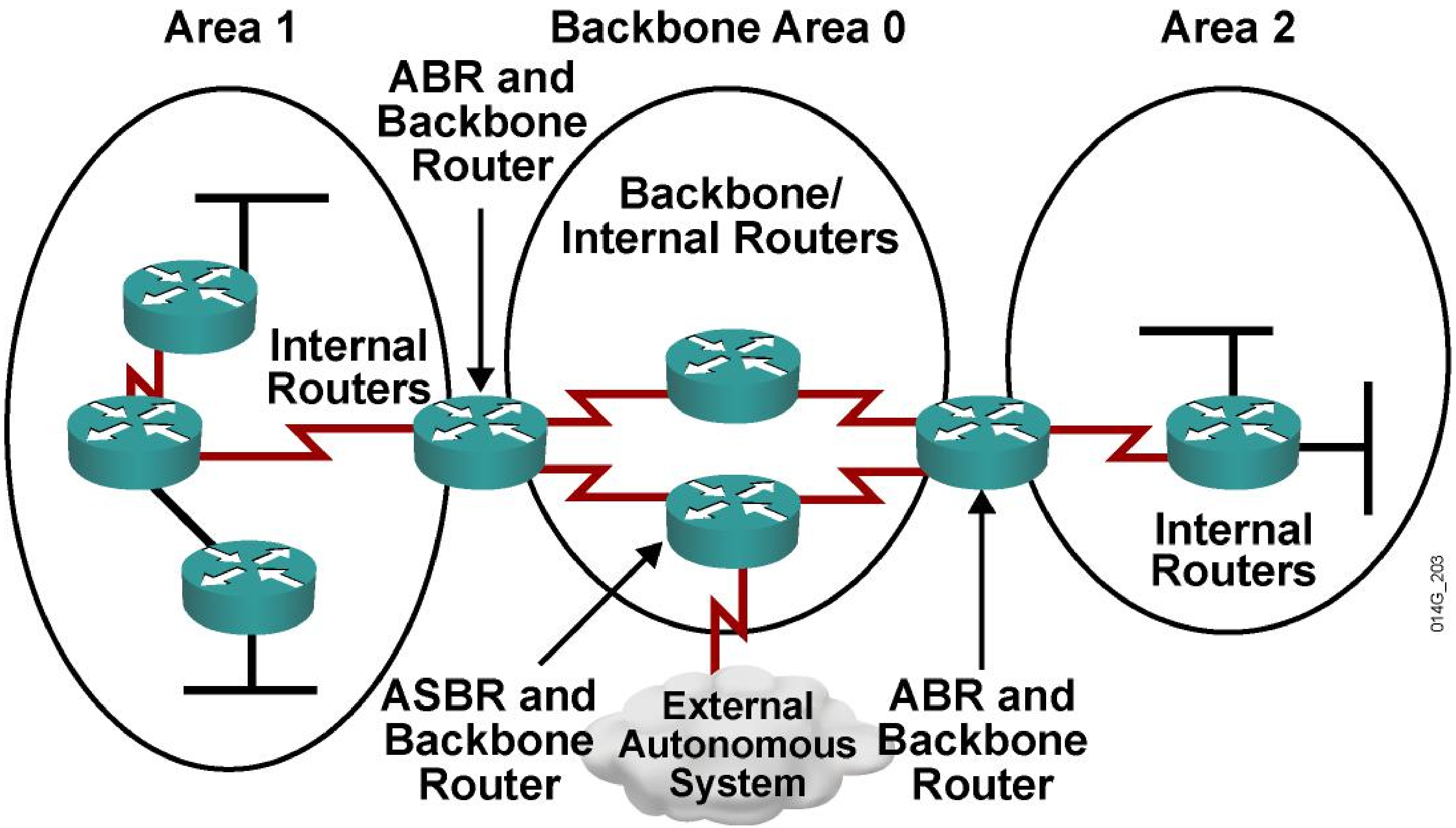


The Solution: OSPF Hierarchical Routing

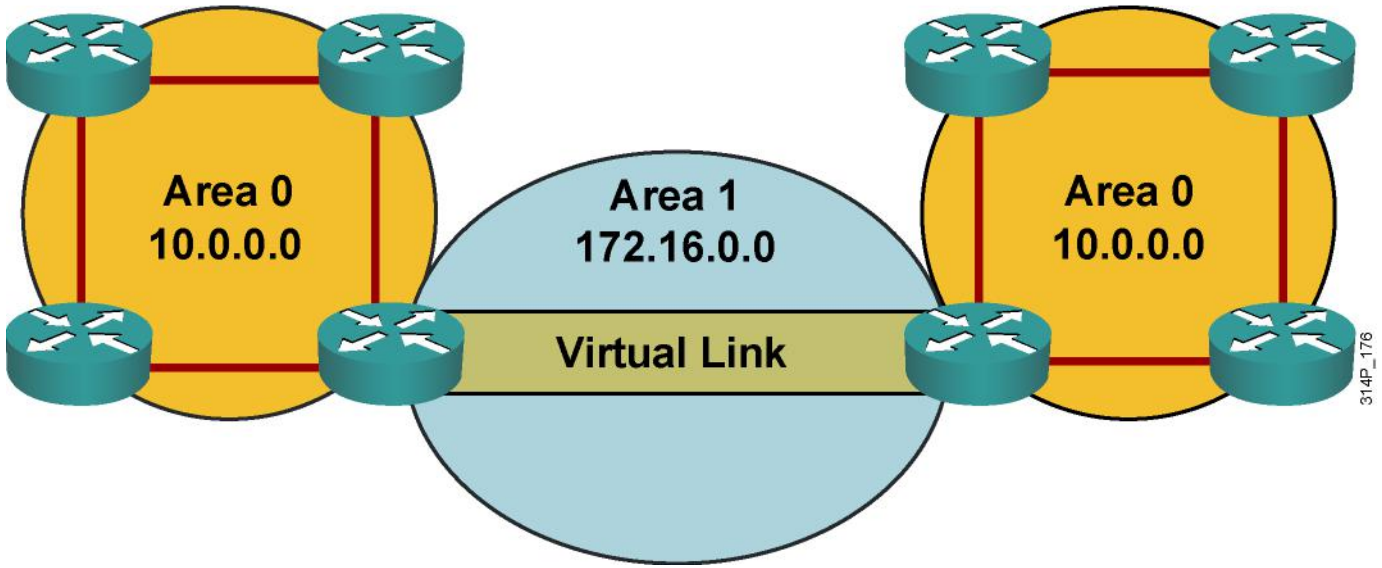
- Consists of areas and autonomous systems
- Minimizes routing update traffic



Types of OSPF Routers



Defining Virtual Links



- **Virtual links are used to connect a discontiguous area to area 0.**
- **A logical connection is built between router A and router B.**
- **Virtual links are recommended for backup or temporary connections.**

Configuring Virtual Links

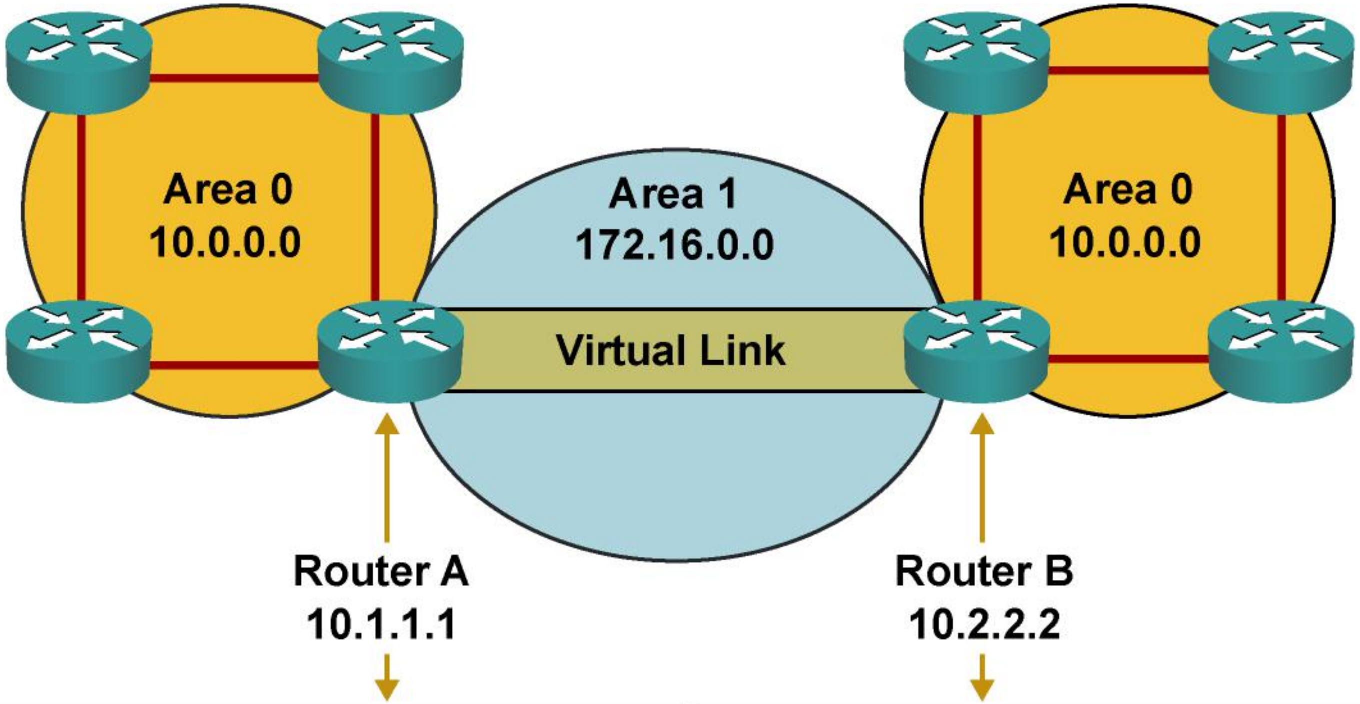
Router(config-router)#

area *area-id* **virtual-link** *router-id* **[authentication [message-digest | null]] [hello-interval** *seconds***] [retransmit-interval** *seconds***] [transmit-delay** *seconds***] [dead-interval** *seconds***] [[authentication-key** ***key*****] | [message-digest-key** ***key-id*** **md5** ***key*****]]**

Creates a virtual link

```
remoterouter#sh ip ospf
 Routing Process "ospf 1000" with ID 10.2.2.2
 Supports only single TOS(TOS0) routes
 Supports opaque LSA
 Supports Link-local Signaling (LLS)
 Supports area transit capability
 It is an area border router
<output omitted>
```

OSPF Virtual Link Configuration Example




```
router ospf 1000
  network 172.16.0.0 0.0.255.255 area 1
  network 10.0.0.0 0.255.255.255 area 0
  area 1 virtual-link 10.2.2.2
```

```
router ospf 1000
  network 172.16.0.0 0.0.255.255 area 1
  network 10.0.0.0 0.255.255.255 area 0
  area 1 virtual-link 10.1.1.1
```

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The show ip ospf virtual-links Command

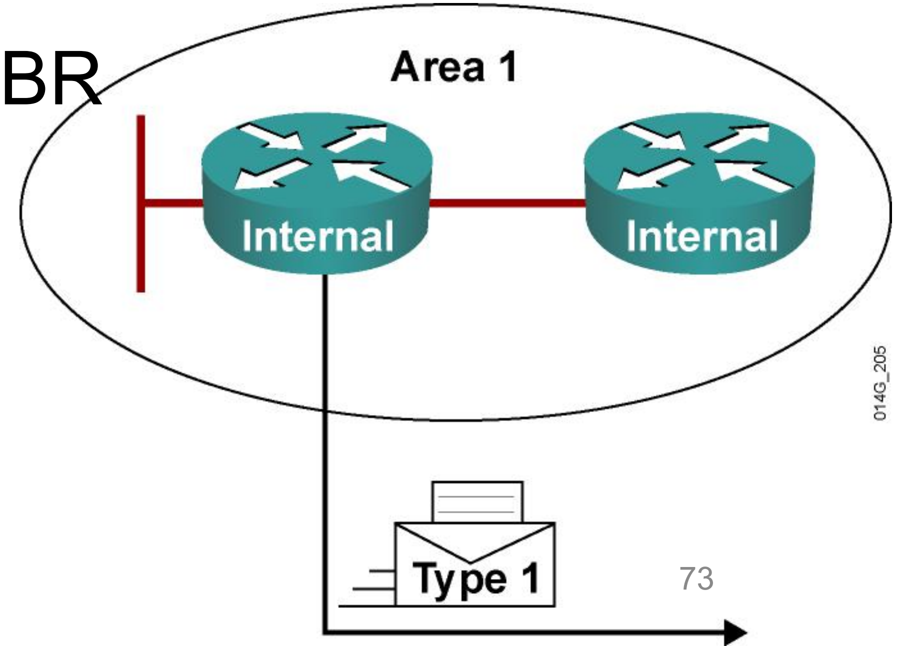
```
RouterA#sh ip ospf virtual-links
Virtual Link OSPF_VL0 to router 10.2.2.2 is up
  Run as demand circuit
  DoNotAge LSA allowed.
  Transit area 1, via interface Serial0/0/1, Cost of using 781
  Transmit Delay is 1 sec, State POINT_TO_POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:07
    Adjacency State FULL (Hello suppressed)
    Index 1/2, retransmission queue length 0, number of retransmission 1
    First 0x0(0)/0x0(0) Next 0x0(0)/0x0(0)
    Last retransmission scan length is 1, maximum is 1
    Last retransmission scan time is 0 msec, maximum is 0 msec
RouterA#
```

LSA Types

LSA Type	Description
1	Router LSAs
2	Network LSAs
3 or 4	Summary LSAs
5	Autonomous system external LSAs
6	Multicast OSPF LSA
7	Defined for not-so-stubby areas
8	External attributes LSA for Border Gateway Protocol (BGP)
9, 10, 11	Opaque LSAs

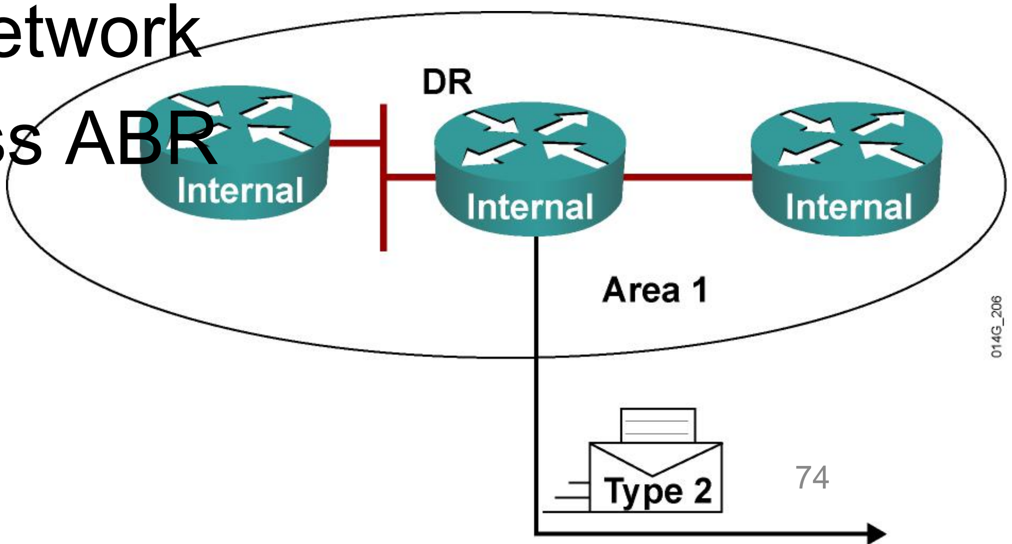
LSA Type 1: Router LSA

- One router LSA (type 1) for every router in an area
  - Includes list of directly attached links
  - Each link identified by IP prefix assigned to link and link type
- Identified by the router ID of the originating router
- Floods within its area only; does not cross ABR

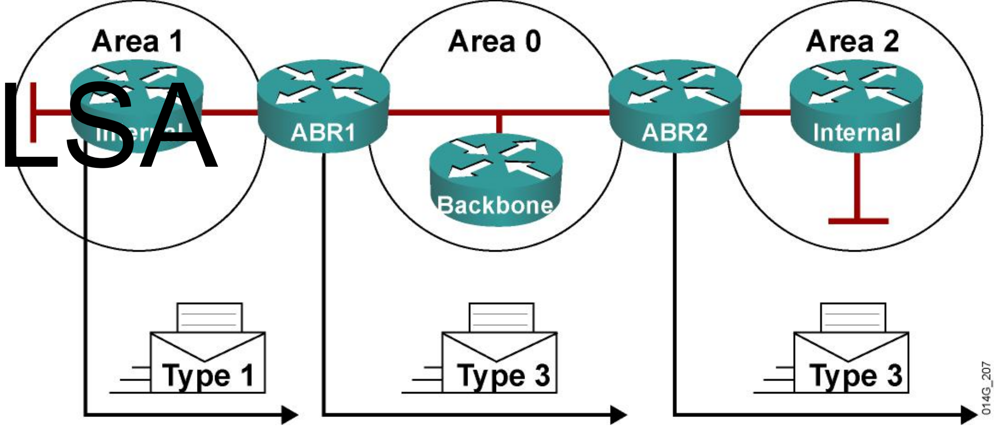


LSA Type 2: Network LSA

- One network (type 2) LSA for each transit broadcast or NBMA network in
  an area
  - Includes list of attached routers on the transit link
  - Includes subnet mask of link
- Advertised by the DR of the broadcast network
- Floods within its area only; does not cross ABR

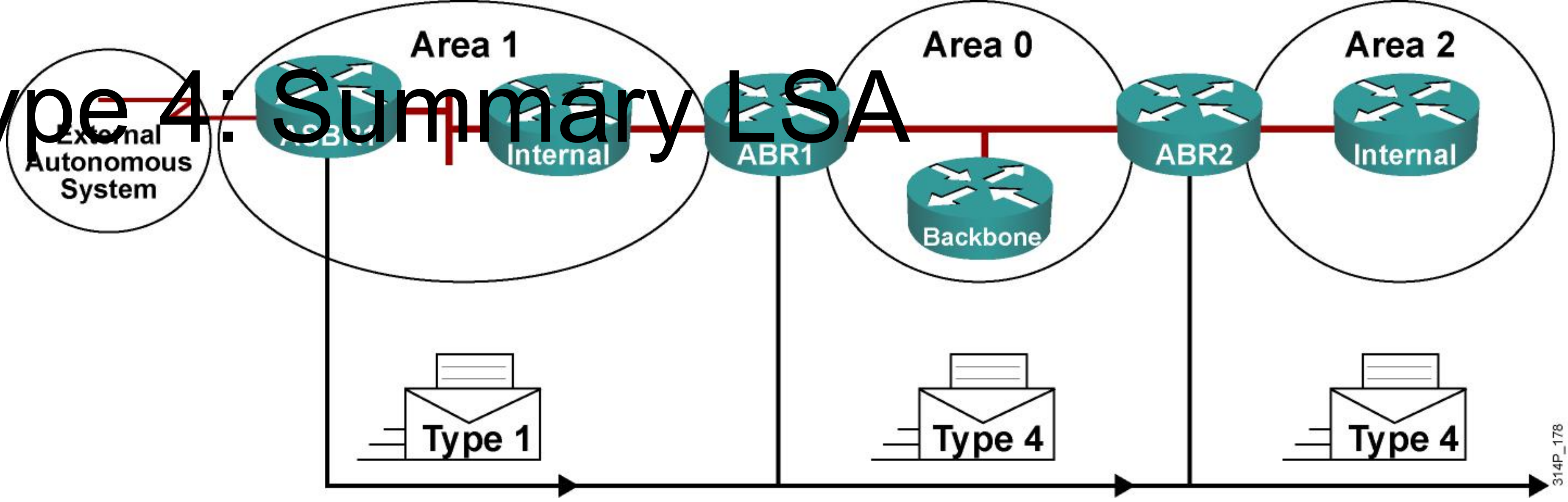


LSA Type 3: Summary LSA



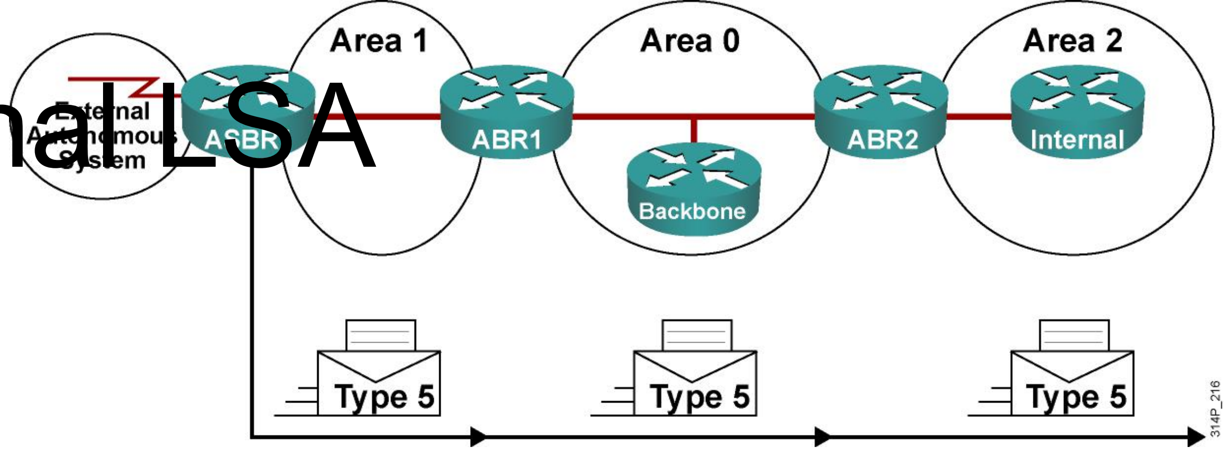
- Type 3 LSAs are used to flood network information to areas outside the originating area (interarea)
  - Describes network number and mask of link.
- Advertised by the ABR of originating area.
- Regenerated by subsequent ABRs to flood throughout the autonomous system.
- By default, routes are not summarized, and type 3 LSA is advertised for every subnet.

LSA Type 4: Summary LSA



- Summary (type 4) LSAs are used to advertise an ASBR to all other areas in the autonomous system.
- They are generated by the ABR of the originating area.
- They are regenerated by all subsequent ABRs to flood throughout the autonomous system.
- Type 4 LSAs contain the router ID of the ASBR.

LSA Type 5: External LSA



- External (type 5) LSAs are used to advertise networks from other autonomous systems.
- Type 5 LSAs are advertised and owned by the originating ASBR.
- Type 5 LSAs flood throughout the entire autonomous system.
- The advertising router ID (ASBR) is unchanged throughout the autonomous system.
- Type 4 LSA is needed to find the ASBR.
- By default, routes are not summarized.

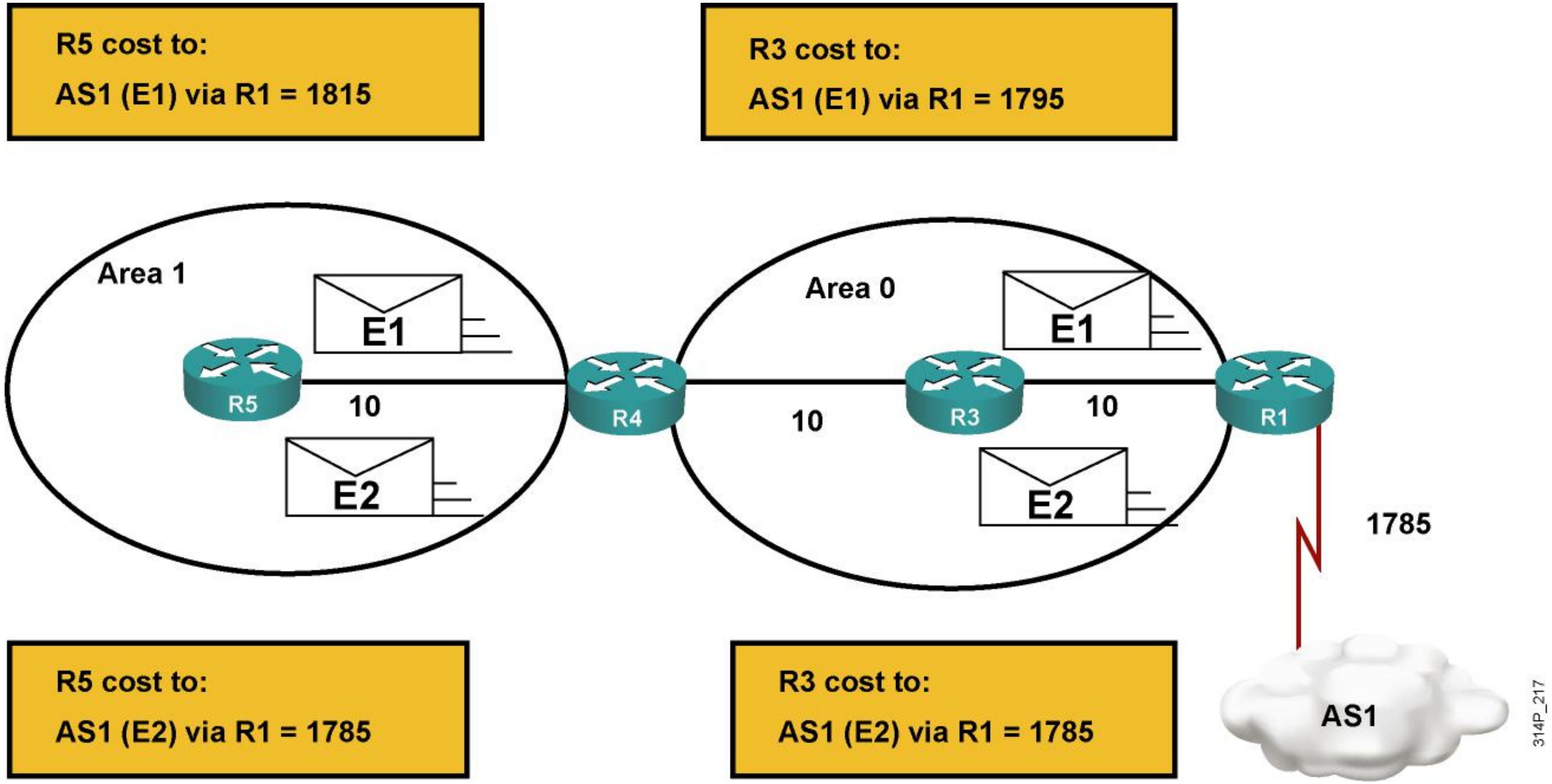
Interpreting the OSPF Database

```
RouterA#show ip ospf database
     OSPF Router with ID (10.0.0.11) (Process ID 1)
                Router Link States (Area 0)
Link ID         ADV Router      Age         Seq#       Checksum Link count
10.0.0.11        10.0.0.11       548         0x80000002 0x00401A 1
10.0.0.12        10.0.0.12       549         0x80000004 0x003A1B 1
100.100.100.100 100.100.100.100 548         0x800002D7 0x00EEA9 2
                Net Link States (Area 0)
Link ID         ADV Router      Age         Seq#       Checksum
172.31.1.3       100.100.100.100 549         0x80000001 0x004EC9
                Summary Net Link States (Area 0)
Link ID         ADV Router      Age         Seq#       Checksum
10.1.0.0         10.0.0.11       654         0x80000001 0x00FB11
10.1.0.0         10.0.0.12       601         0x80000001 0x00F516
<output omitted>
```

Interpreting the Routing Table: Types of Routes

Router Designator		Description
O	OSPF intra-area (router LSA) and network LSA	• Networks from within the area of the router • Advertised by way of router LSAs and network LSA
O IA	OSPF interarea (summary LSA)	• Networks from outside the area of the router, but within the OSPF autonomous system • Advertised by way of summary LSAs
O E1	Type 1 external routes	• Networks outside of the autonomous system of the router • Advertised by way of external LSAs
O E2	Type 2 external routes	

Calculating Costs for E1 and E2 Routes



The show ip route Command

```
RouterB>show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

     172.31.0.0/24 is subnetted, 2 subnets
O IA    172.31.2.0 [110/1563] via 10.1.1.1, 00:12:35, FastEthernet0/0
O IA    172.31.1.0 [110/782] via 10.1.1.1, 00:12:35, FastEthernet0/0
     10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
C       10.200.200.13/32 is directly connected, Loopback0
C       10.1.3.0/24 is directly connected, Serial0/0/0
O       10.1.2.0/24 [110/782] via 10.1.3.4, 00:12:35, Serial0/0/0
C       10.1.1.0/24 is directly connected, FastEthernet0/0
O       10.1.0.0/24 [110/782] via 10.1.1.1, 00:12:37, FastEthernet0/0
O E2    10.254.0.0/24 [110/50] via 10.1.1.1, 00:12:37, FastEthernet0/0
```

OSPF LSDB Overload Protection

Router(config-router)#

max-lsa *maximum-number* **[***threshold-percentage***] [warning-only] [ignore-time** minutes**] [ignore-count** *count-number***] [reset-time** *minutes***]**

- Excessive LSAs generated by other routers can drain local router resources.
- This feature can limit the processing of non-self-generated LSAs for a defined OSPF process.

Changing the Cost Metric

- Dijkstra's algorithm determines the best path by adding all link costs along a path.
- The cost, or metric, is an indication of the overhead to send packets over an interface. Default = (100 Mbps) / (bandwidth in Mbps).

RouterA(config-if)#

ip ospf cost *interface-cost*

- Overrides the default cost calculation. Values from 1 to 65535 can be defined.

RouterA(config-router)#

auto-cost reference-bandwidth *ref-bw*

- Sets the reference bandwidth to values other than 100 Mbps (legal values range from 1 to 4,294,967 in megabits per second).

Summary

- There are four OSPF router types: internal routers, backbone routers, ABRs, and ASBRs.
- A virtual link allows discontiguous area 0’ to be connected, or a disconnected area to be connected to area 0, via a transit area. Virtual links should be used only for temporary connections or backup after a failure, not as a primary backbone design feature.
- There are 11 OSPF LSA types. The first five are the most commonly used:
  - Type 1 router
  - Type 2 network
  - Type 3 and 4 summary
  - Type 5 external

Summary (Cont.)

- In the IP routing table, OSPF routes are classified as either intra-area, interarea, or external; external routes are subdivided into E1 and E2.

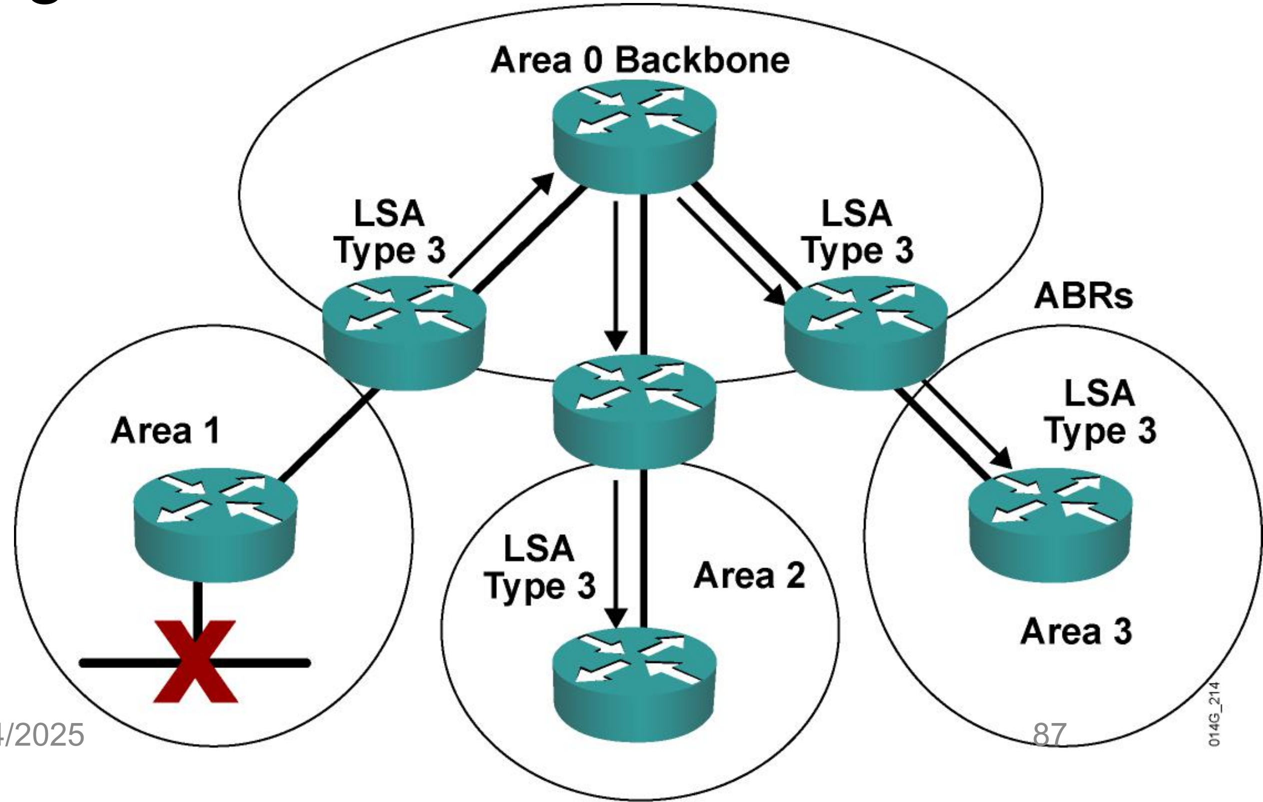
- OSPF LSDB overload protection limits the processing of non-self-generated LSAs.
- The OSPF cost defaults to (100 Mbps) / (bandwidth in megabits per second). The cost can be changed on a per-interface basis, and the reference bandwidth (100 Mbps) can also be changed.

Configuring OSPF

OSPF Route Summarization

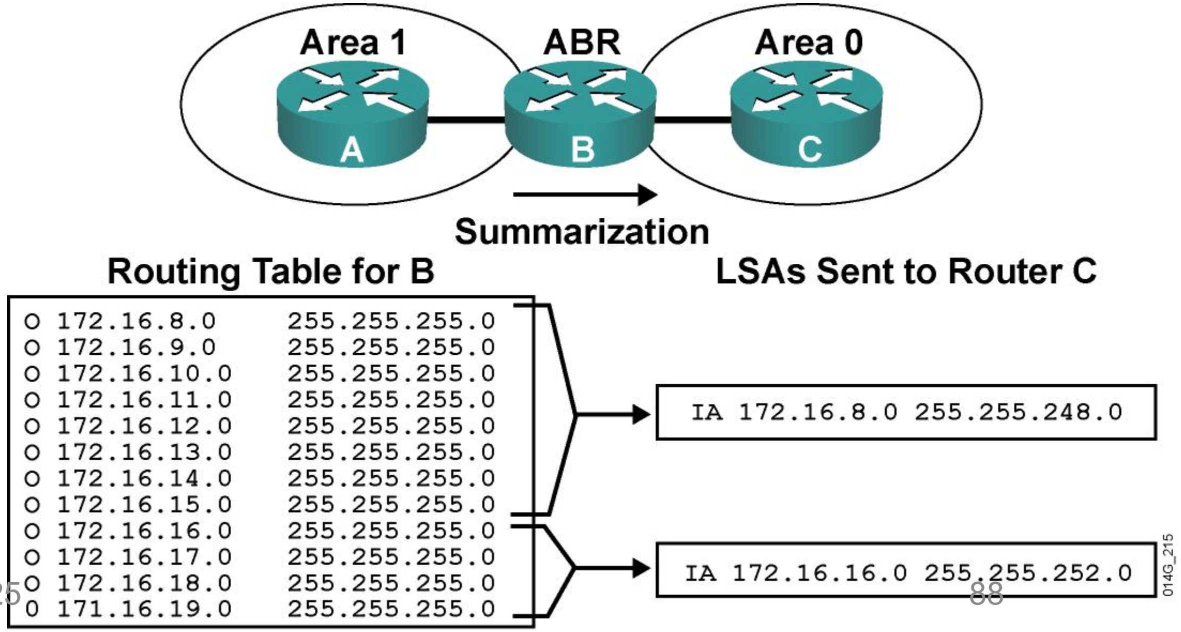
Benefits of Route Summarization

- Minimizes number of routing table entries
- Localizes impact of a topology change
- Reduces LSA type 3 and 5 flooding and saves CPU resources



Using Route Summarization

- Interarea summary link carries mask.
- One or more entries can represent several subnets.



Configuring Route Summarization

Router(config-router)#

area *area-id* range *address mask* [advertise | not-advertise] [cost *cost*]

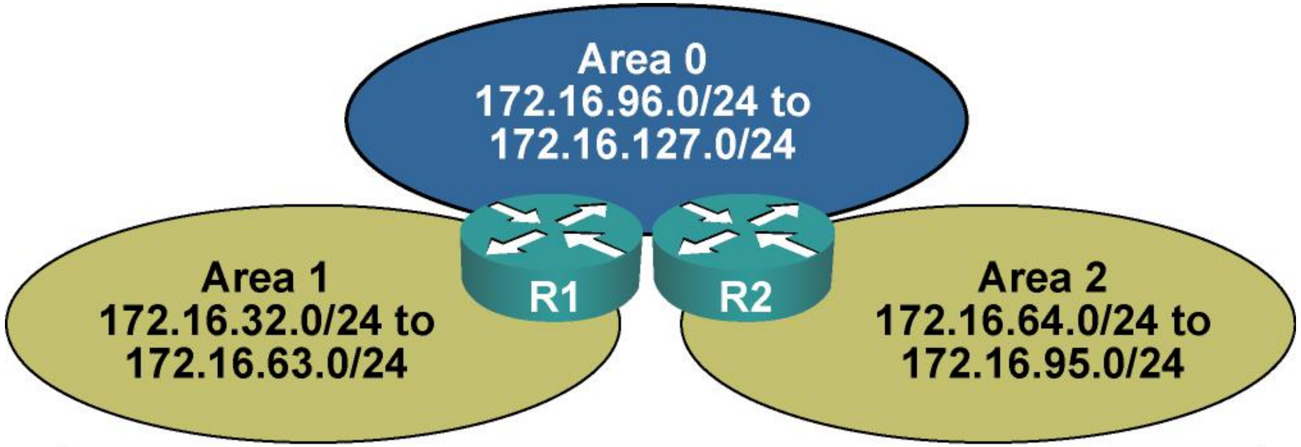
- **Consolidates interarea routes on an ABR**

Router(config-router)#

summary-address *ip-address mask* [not-advertise] [tag *tag*]

- **Consolidates external routes, usually on an ASBR**

Route Summarization Configuration Example at ABR




```
Router1(config)# router ospf 100
Router1(config-router)#network 172.16.32.1 0.0.0.0 area 1
Router1(config-router)#network 172.16.96.1 0.0.0.0 area 0
Router1(config-router)#area 0 range 172.16.96.0 255.255.224.0
Router1(config-router)#area 1 range 172.16.32.0 255.255.224.0
```

```
Router2(config)# router ospf 100
Router2(config-router)#network 172.16.64.1 0.0.0.0 area 2
Router2(config-router)#network 172.16.127.1 0.0.0.0 area 0
Router2(config-router)#area 0 range 172.16.96.0 255.255.224.0
Router2(config-router)#area 2 range 172.16.64.0 255.255.224.0
```

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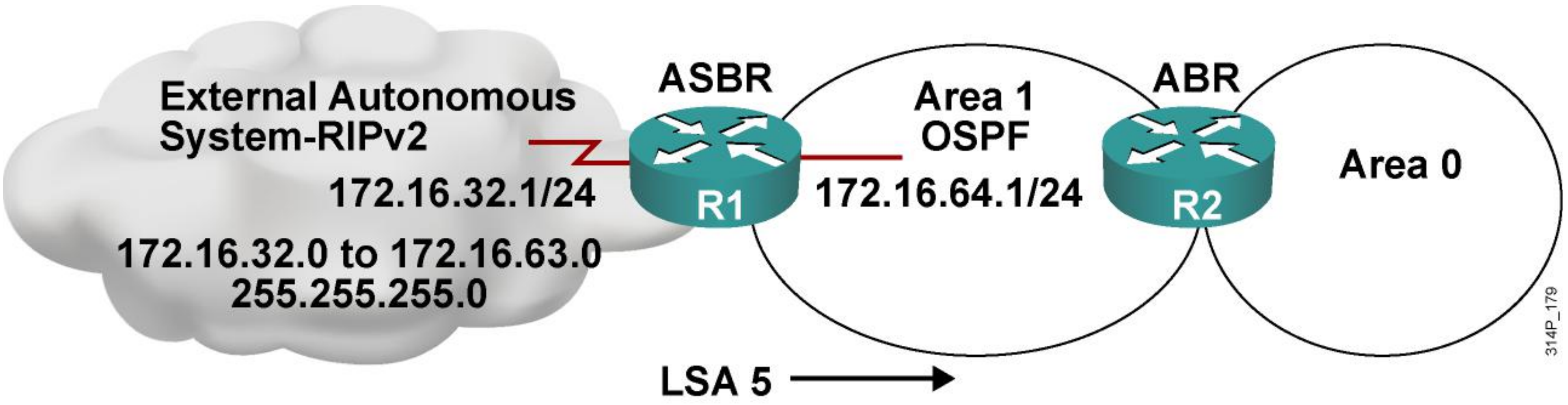
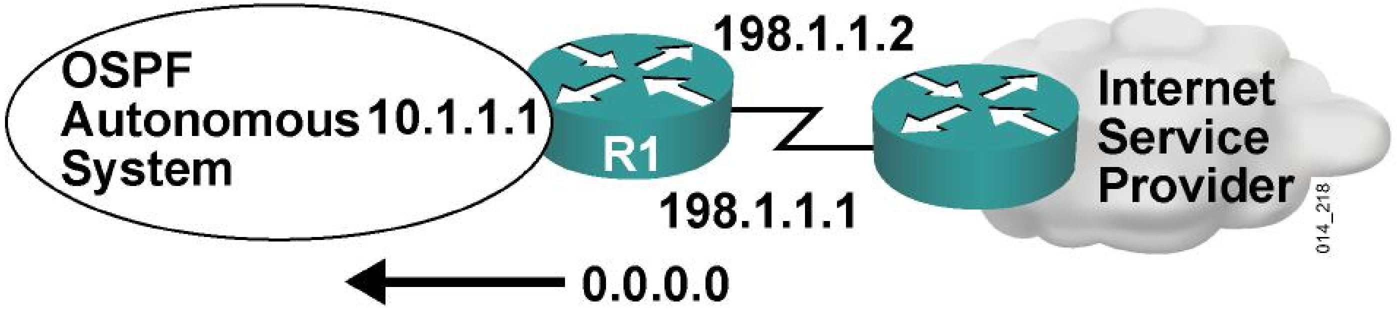
Route Summarization Configuration Example at ASBR




```
R1#
router ospf 100
network 172.16.64.1  0.0.0.0 area 1
summary-address 172.16.32.0 255.255.224.0
```

Default Routes in OSPF



- A default route is injected into OSPF as an external LSA type 5.
- Default route distribution is not on by default; use the default-information originate command under the OSPF routing process.

Configuring OSPF Default Routes

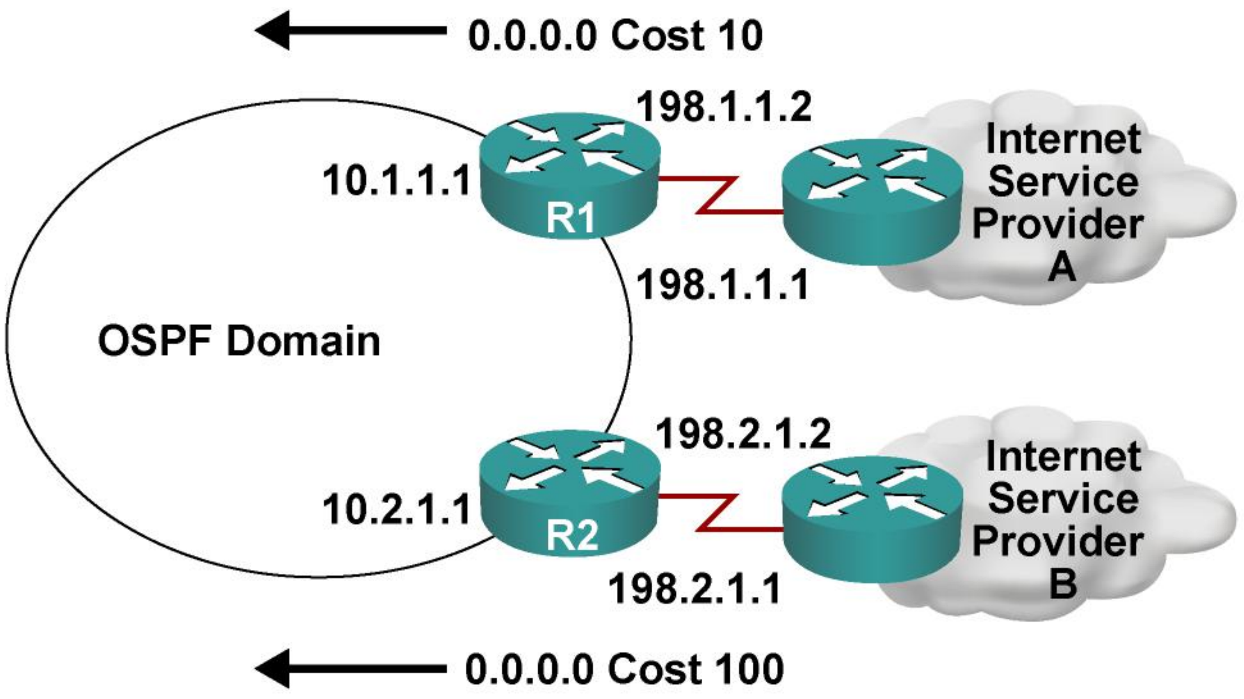
Router(config-router)#

default-information originate [always] [metric *metric-value***]**

[metric-type *type-value***] [route-map** *map-name***]**

- **Normally, this command advertises a 0.0.0.0 default into the OSPF network only if the default route already exists in the routing table.**
- **The** always **keyword allows the 0.0.0.0 default to be advertised even when the default route does not exist in the routing table.**

Default Route Configuration Example




```
R1#
router ospf 100
network 10.1.1.1  0.0.0.0 area 0
default-information originate metric 10

ip route 0.0.0.0 0.0.0.0 198.1.1.2
```

```
R2#
router ospf 100
network 10.2.1.1  0.0.0.0 area 0
default-information originate metric 100

ip route 0.0.0.0 0.0.0.0 198.2.1.2
```

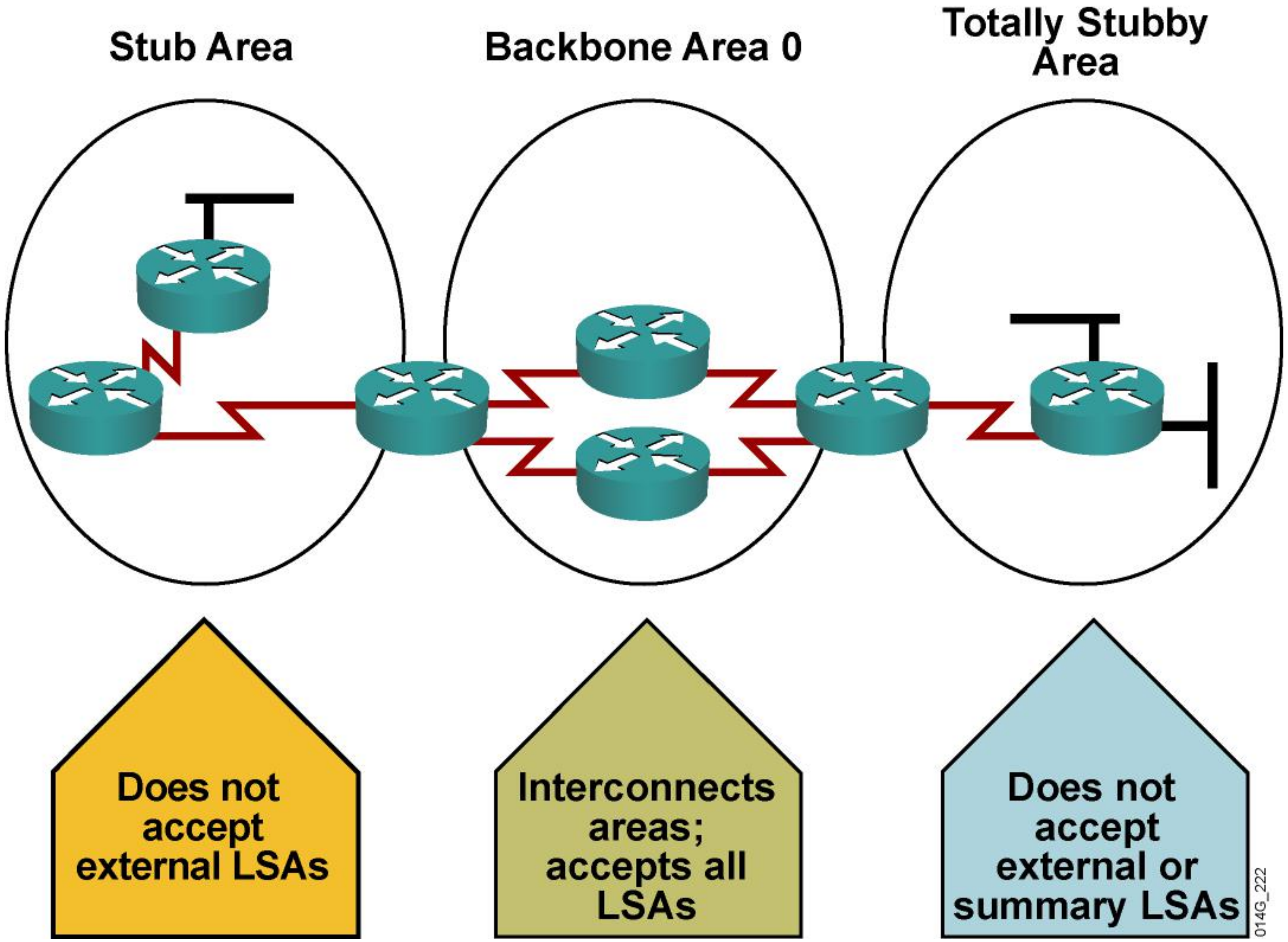
Summary

- Route summarization improves CPU utilization, reduces LSA flooding, and reduces routing table sizes.
- The area range command is used to summarize at the ABR. The summary-address command is used to summarize at the ASBR.
- Default routes can be used in OSPF to prevent the need for a specific route to all destination networks. The benefits include a much smaller routing table and LSDB, with complete reachability.
- OSPF uses the default-information originate command to inject a default route.

Configuring OSPF

Configuring OSPF Special Area Types

Types of Areas

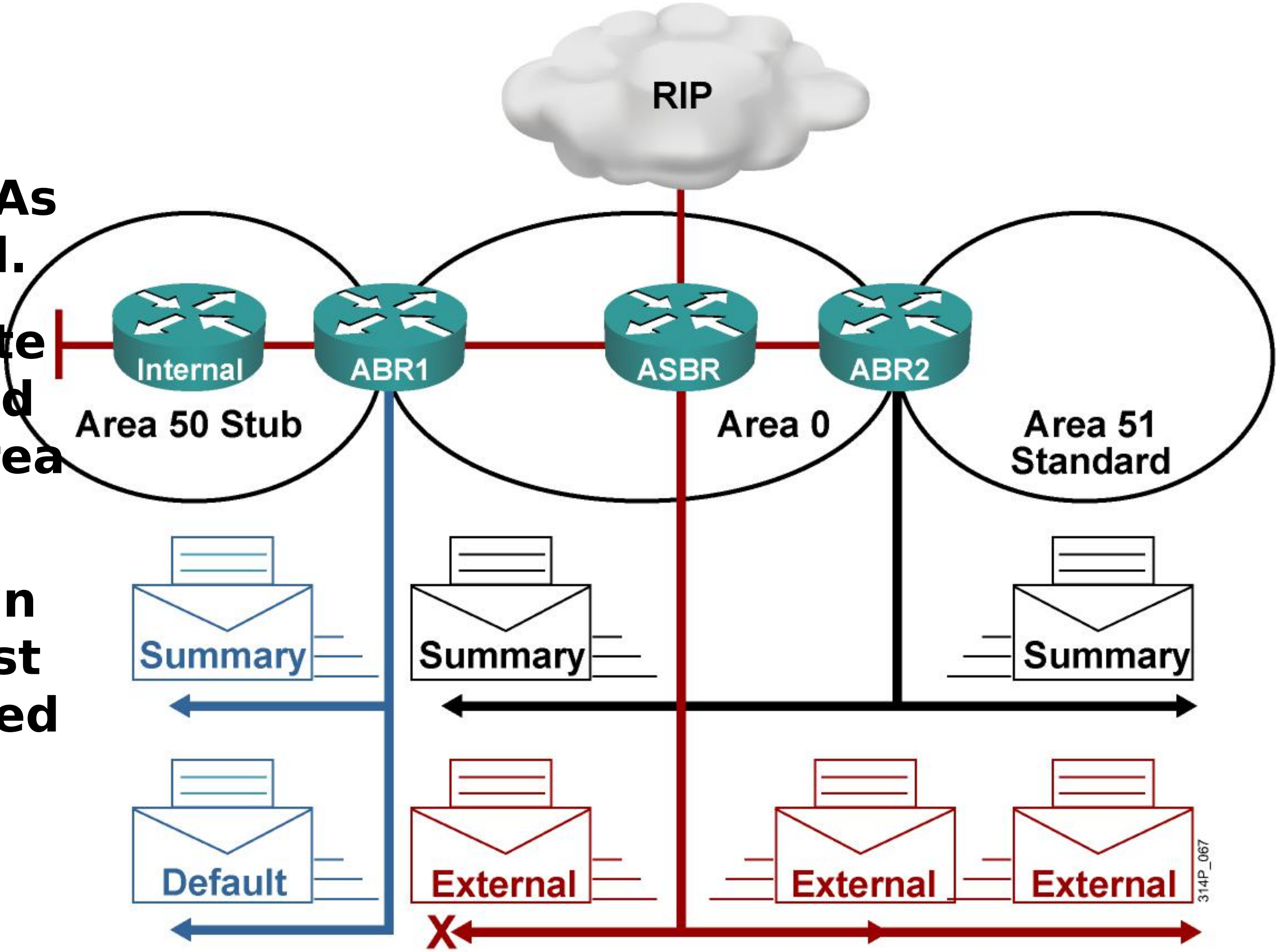


Stub and Totally Stub Area Rules

- An area can be stub or totally stub if:
  - There is a single ABR, or if there is more than one ABR, suboptimal routing paths to other areas or external autonomous systems are acceptable.
  - All routers in the area are configured as stub routers.
  - There is no ASBR in the area.
  - The area is not area 0.
  - No virtual links go through the area.

Using Stub Areas

- **External LSAs are stopped.**
- **Default route is advertised into stub area by the ABR.**
- **All routers in area 50 must be configured as stub.**



Stub Area Configuration

RouterA(config-router)#

area *area-id* **stub [no-summary]**

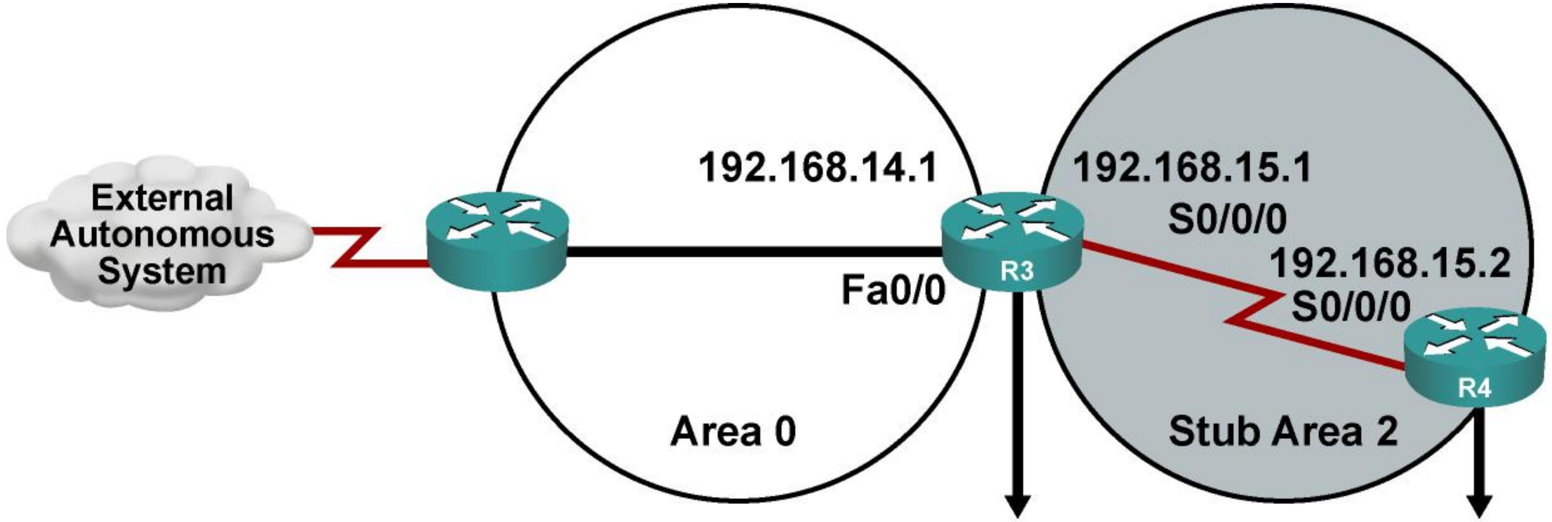
- **This command turns on stub area networking.**
- **All routers in a stub area must use the** stub **command.**

RouterA(config-router)#

area *area-id* **default-cost** *cost*

- **This command defines the cost of a default route sent into the stub area.**
- **The default cost is 1.**

OSPF Stub Area Configuration Example



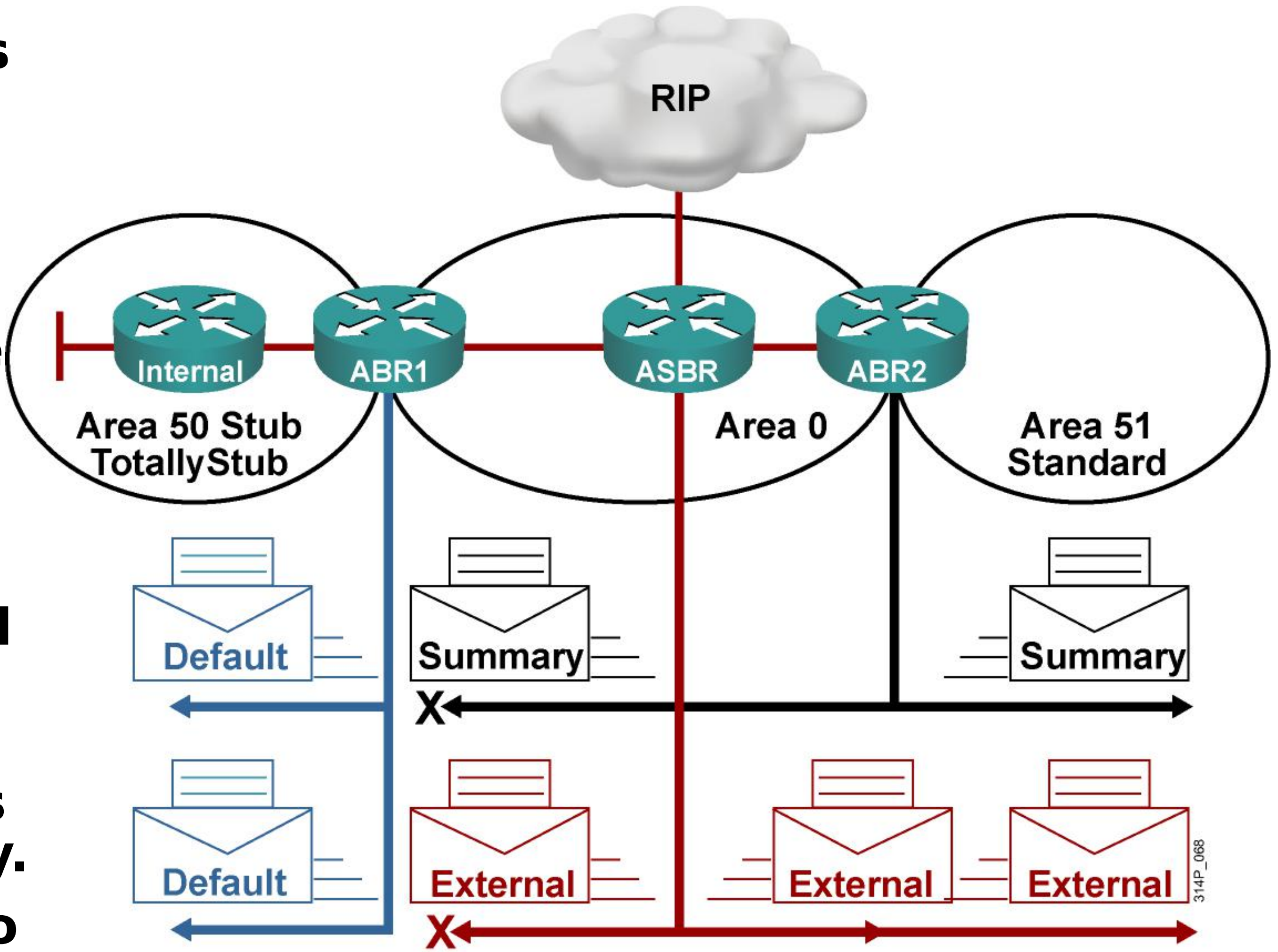

```
R3#

interface FastEthernet 0/0
ip address 192.168.14.1 255.255.255.0
interface Serial 0/0/0
ip address 192.168.15.1 255.255.255.252

router ospf 100
network 192.168.14.0 0.0.0.255 area 0
network 192.168.15.0 0.0.0.255 area 2
area 2 stub
```

```
R4#

interface Serial 0/0/0
ip address 192.168.15.2 255.255.255.252

router ospf 15
network 192.168.15.0 0.0.0.255 area 2
area 2 stub
```

314P_180

Using Totally Stubby Areas

- **External LSAs are stopped.**
- **Summary LSAs are stopped.**
- **Routing table is reduced to a minimum.**
- **All routers must be configured as stub.**
- **ABR must be configured as totally stubby.**
- **This is a Cisco proprietary feature.**



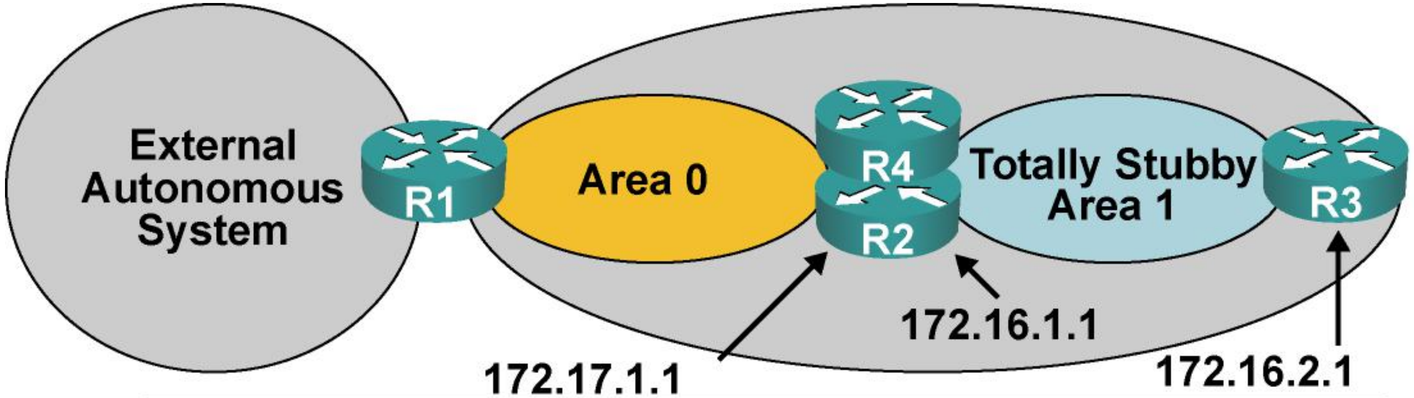
Totally Stubby Configuration

RouterA(config-router)#

area *area-id* **stub no-summary**

- **The addition of** no-summary **on the ABR creates a totally stubby area and prevents all summary LSAs from entering the stub area.**

Totally Stubby Configuration Example



```
Router2(config)# router ospf 10
Router2(config-router)# network 172.17.0.0 0.0.255.255 area 0
Router2(config-router)# network 172.16.0.0 0.0.255.255 area 1
Router2(config-router)#area 1 stub no-summary
Router2(config-router)#area 1 default-cost 5
Router2(config-router)# ! R2 is the preferred ABR
```

```
Router3(config)# router ospf 10
Router3(config-router)# network 172.16.0.0 0.0.255.255 area 1
Router3(config-router)# area 1 stub
```

```
Router4(config)# router ospf 10
Router4(config-router)# network 172.17.0.0 0.0.255.255 area 0
Router4(config-router)# network 172.16.0.0 0.0.255.255 area 1
Router4(config-router)#area 1 stub no-summary
Router4(config-router)#area 1 default-cost 10
```

314P_069

Routing Table in a Standard Area

```
P1R3#sh ip route
<output omitted>

Gateway of last resort is not set
     172.31.0.0/32 is subnetted, 4 subnets
O IA    172.31.22.4 [110/782] via 10.1.1.1, 00:02:44, FastEthernet0/0
O IA    172.31.11.1 [110/1] via 10.1.1.1, 00:02:44, FastEthernet0/0
O IA    172.31.11.2 [110/782] via 10.1.3.4, 00:02:52, Serial0/0/0
                    [110/782] via 10.1.1.1, 00:02:52, FastEthernet0/0
O IA    172.31.11.4 [110/782] via 10.1.1.1, 00:02:44, FastEthernet0/0
     10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
O       10.11.0.0/24 [110/782] via 10.1.1.1, 00:03:22, FastEthernet0/0
C       10.200.200.13/32 is directly connected, Loopback0
C       10.1.3.0/24 is directly connected, Serial0/0/0
O       10.1.2.0/24 [110/782] via 10.1.3.4, 00:03:23, Serial0/0/0
C       10.1.1.0/24 is directly connected, FastEthernet0/0
O       10.1.0.0/24 [110/782] via 10.1.1.1, 00:03:23, FastEthernet0/0
O E2    10.254.0.0/24 [110/50] via 10.1.1.1, 00:02:39, FastEthernet0/0
P1R3#
```

Routing Table in a Stub Area

```
P1R3#sh ip route
<output omitted>

Gateway of last resort is 10.1.1.1 to network 0.0.0.0
     172.31.0.0/32 is subnetted, 4 subnets
O IA    172.31.22.4 [110/782] via 10.1.1.1, 00:01:49, FastEthernet0/0
O IA    172.31.11.1 [110/1] via 10.1.1.1, 00:01:49, FastEthernet0/0
O IA    172.31.11.2 [110/782] via 10.1.3.4, 00:01:49, Serial0/0/0
                    [110/782] via 10.1.1.1, 00:01:49, FastEthernet0/0
O IA    172.31.11.4 [110/782] via 10.1.1.1, 00:01:49, FastEthernet0/0
     10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
O       10.11.0.0/24 [110/782] via 10.1.1.1, 00:01:50, FastEthernet0/0
C       10.200.200.13/32 is directly connected, Loopback0
C       10.1.3.0/24 is directly connected, Serial0/0/0
O       10.1.2.0/24 [110/782] via 10.1.3.4, 00:01:50, Serial0/0/0
C       10.1.1.0/24 is directly connected, FastEthernet0/0
O       10.1.0.0/24 [110/782] via 10.1.1.1, 00:01:50, FastEthernet0/0
O*IA 0.0.0.0/0 [110/2] via 10.1.1.1, 00:01:51, FastEthernet0/0
P1R3#
```

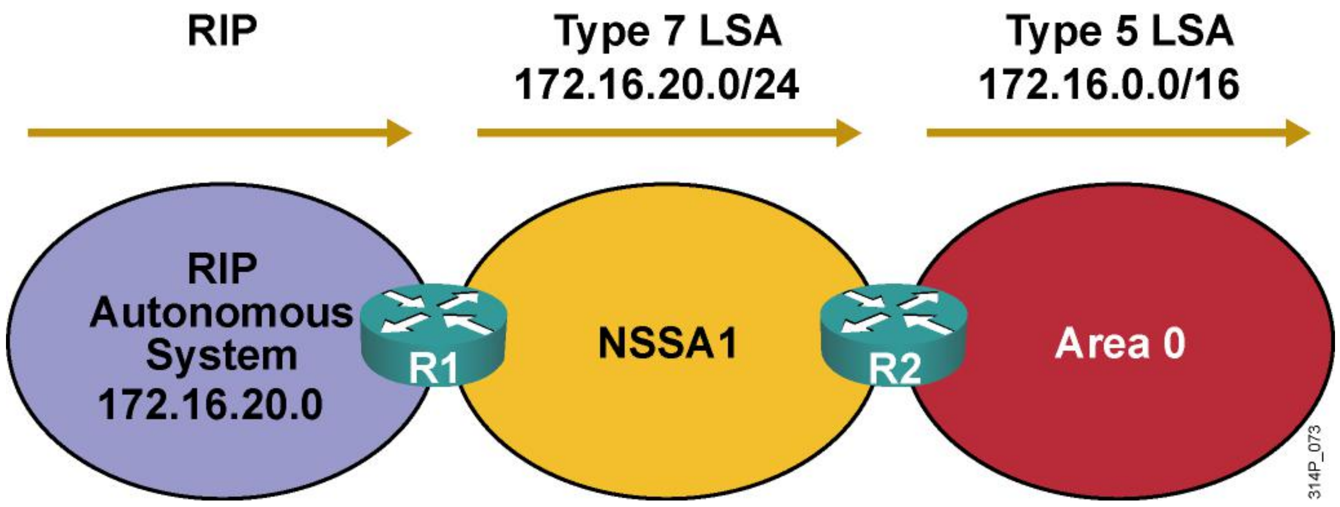
Routing Table in a Stub Area with Summarization

```
P1R3#sh ip route
<output omitted>

Gateway of last resort is 10.1.1.1 to network 0.0.0.0
     172.31.0.0/16 is variably subnetted, 2 subnets, 2 masks
O IA    172.31.22.4/32 [110/782] via 10.1.1.1, 00:13:08, FastEthernet0/0
O IA    172.31.11.0/24 [110/1] via 10.1.1.1, 00:02:39, FastEthernet0/0
     10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
O       10.11.0.0/24 [110/782] via 10.1.1.1, 00:13:08, FastEthernet0/0
C       10.200.200.13/32 is directly connected, Loopback0
C       10.1.3.0/24 is directly connected, Serial0/0/0
O       10.1.2.0/24 [110/782] via 10.1.3.4, 00:13:09, Serial0/0/0
C       10.1.1.0/24 is directly connected, FastEthernet0/0
O       10.1.0.0/24 [110/782] via 10.1.1.1, 00:13:09, FastEthernet0/0
O*IA 0.0.0.0/0 [110/2] via 10.1.1.1, 00:13:09, FastEthernet0/0
P1R3#
```

Routing Table in a Totally Stubby Area

```
P1R3#sh ip route
<output omitted>

Gateway of last resort is 10.1.1.1 to network 0.0.0.0
     10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
O       10.11.0.0/24 [110/782] via 10.1.1.1, 00:16:53, FastEthernet0/0
C       10.200.200.13/32 is directly connected, Loopback0
C       10.1.3.0/24 is directly connected, Serial0/0/0
O       10.1.2.0/24 [110/782] via 10.1.3.4, 00:16:53, Serial0/0/0
C       10.1.1.0/24 is directly connected, FastEthernet0/0
O       10.1.0.0/24 [110/782] via 10.1.1.1, 00:16:53, FastEthernet0/0
O*IA 0.0.0.0/0 [110/2] via 10.1.1.1, 00:00:48, FastEthernet0/0
P1R3#
```

Not-So-Stubby Areas



- **NSSA breaks stub area rules.**
- **ASBR (R1) is allowed in NSSA.**
- **Special LSA type 7 defined, sent by ASBR.**
- **ABR (R2) converts LSA type 7**
  to LSA type 5.
- **ABR sends default route into NSSA instead of external routes from other ASBRs.**
- **NSSA is an RFC**

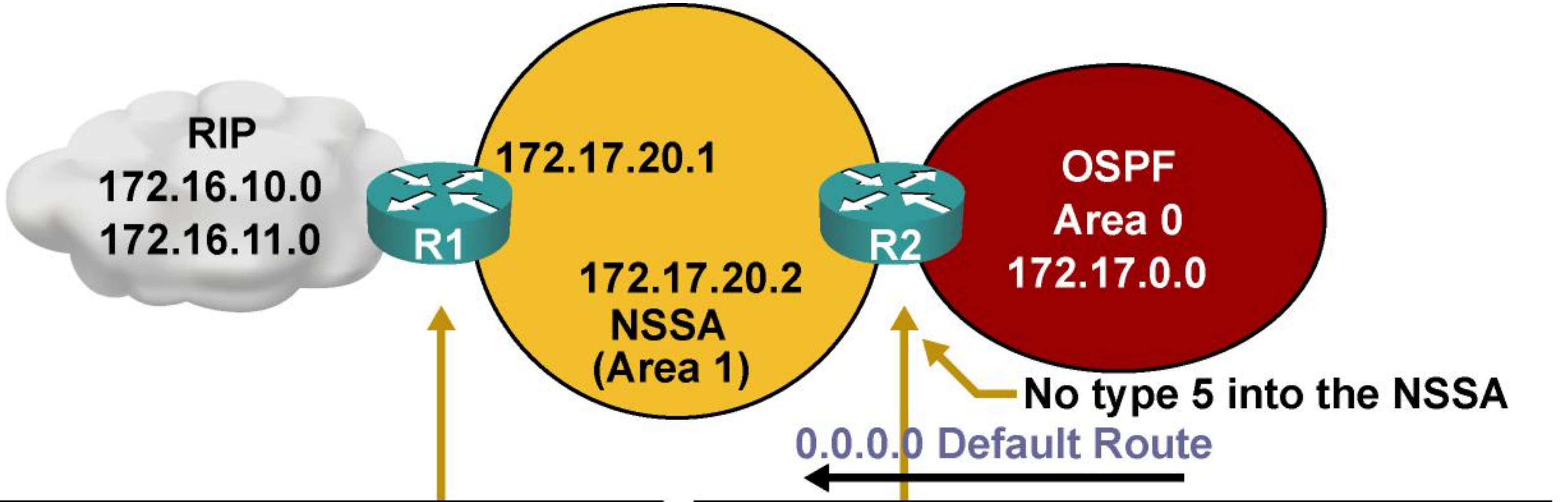
NSSA Configuration

RouterA(config-router)#

area *area-id* **nssa [no-redistribution] [default-information-originate [metric** *metric-value***] [metric-type** *type-value***]] [no-summary]**

- **Use this command instead of the** area stub **command to define the area as NSSA.**
- **The** no-summary **keyword creates an NSSA totally stubby area; this is a Cisco proprietary feature.**

Example: NSSA Configuration

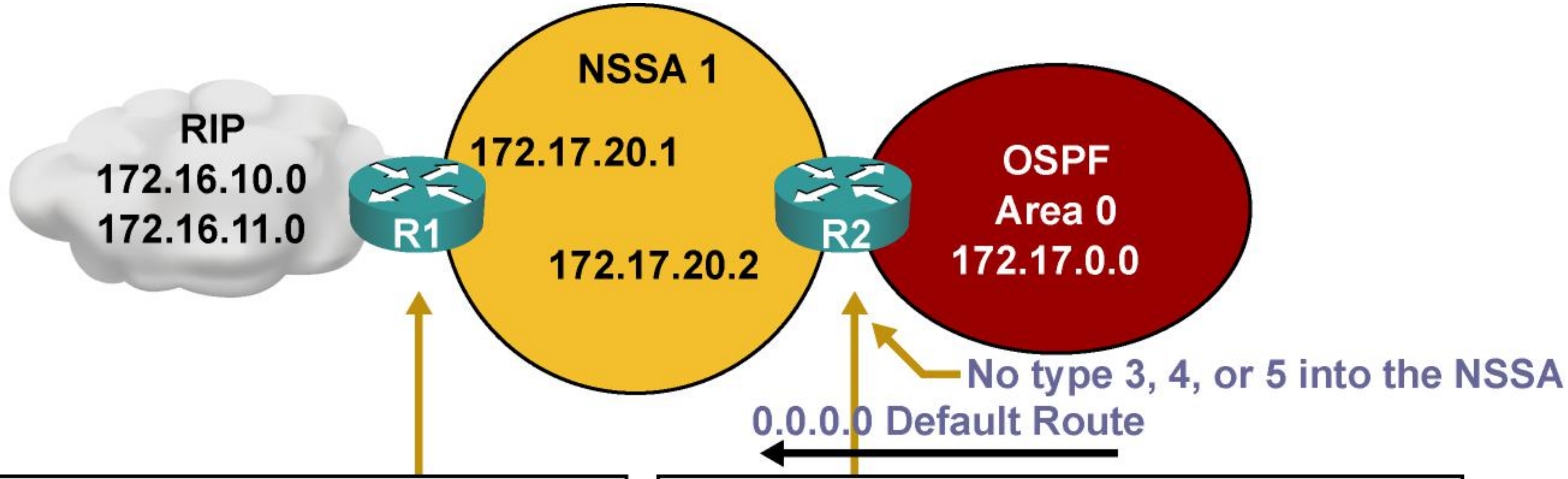



```
R1:
router ospf 10
  redistribute rip subnets
  default metric 150
  network 172.17.0.0 0.0.255.255 area 1
  area 1 nssa
```

```
R2:
router ospf 10
  summary-address 172.16.0.0 255.255.0.0
  network 172.17.20.0 0.0.0.255 area 1
  network 172.17.0.0 0.0.255.255 area 0
  area 1 nssa default-information-originate
```

314P_074

NSSA Totally Stubby Configuration




```
R1:
router ospf 10
  redistribute rip subnets
  default metric 150
  network 172.17.0.0 0.0.255.255 area 1
  area 1 nssa
```

```
R2:
router ospf 10
  summary-address 172.16.0.0 255.255.0.0
  network 172.17.20.0 0.0.0.255 area 1
  network 172.17.0.0 0.0.255.255 area 0
  area 1 nssa no-summary
```

314P_075

- **NSSA totally stubby area is a Cisco proprietary feature.**

show Commands for Stub and NSSA

RouterA#

```
show ip ospf
```

- **Displays which areas are normal, stub, or NSSA**

RouterA#

```
show ip ospf database
```

- **Displays details of LSAs**

RouterA#

```
show ip ospf database nssa-external
```

- **Displays specific details of each LSA type 7 update in database**

RouterA#

```
show ip route
```

- **Displays all routes**

Summary

- There are several OSPF area types: standard, backbone, stub, totally stubby, and NSSA.
- Use the area area-id stub command to define an area as stubby.
- Use the area area-id stub command with the no-summary keyword on the ABR only to define an area as totally stubby.
- For stub areas, external routes are not visible in the routing table, but are accessible via the intra-area default route. For totally stubby areas, interarea and external routes are not visible in the routing table, but are accessible via the intra-area default route.
- Use the area area-id nssa command to define an area as NSSA.
- Use show ip ospf, show ip ospf database, show ip route commands to verify all types of stub areas. Use the show ip ospf database nssa-external command to display details of type 7 LSAs.

Configuring OSPF

Configuring OSPF Authentication

OSPF Authentication Types

- OSPF supports 2 types of authentication:
  - Simple password (or plain text) authentication
  - MD5 authentication
- Router generates and checks every OSPF packet. Router authenticates the source of each routing update packet that it receives.
- Configure a “key” (password); each participating neighbor must have same key configured.

Configuring OSPF Simple Password Authentication

Router(config-if)#

```
ip ospf authentication-key password
```

- **Assigns a password to be used with neighboring routers**

Router(config-if)#

```
ip ospf authentication [message-digest | null]
```

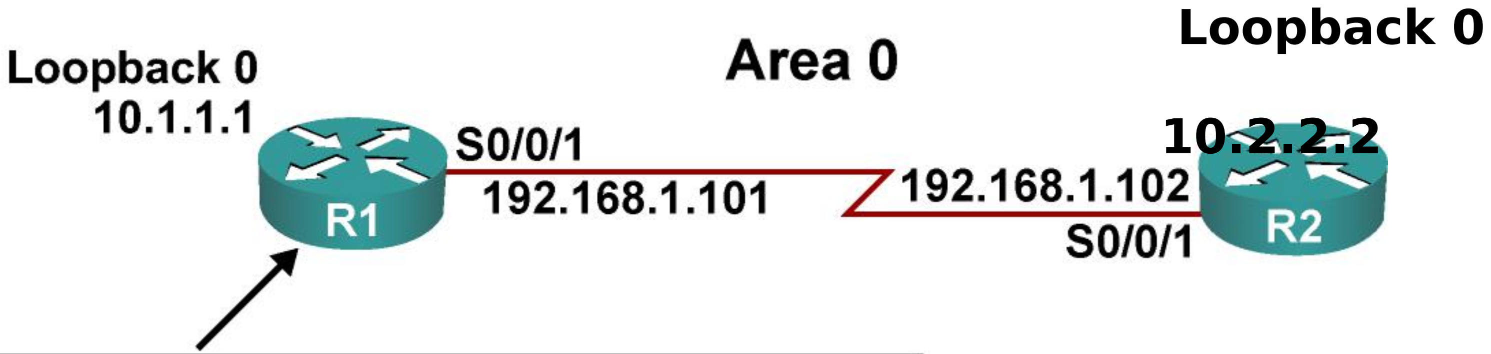
- **Specifies the authentication type for an interface (since Cisco IOS software 12.0)**

Router(config-router)#

```
area area-id authentication [message-digest]
```

- **Specifies the authentication type for an area (was in Cisco IOS software before 12.0)**

Example Simple Password Authentication Configuration




```
<output omitted>
interface Loopback0
 ip address 10.1.1.1. 255.255.255.0

<output omitted>
interface Serial0/0/1
 ip address 192.168.1.101 255.255.255.224
 ip ospf authentication
 ip ospf authetication-key plainpas

<output omitted>
router ospf 10
 log-adjacency-changes
 network 10.1.1.1 0.0.0.0 area 0
 network 192.168.1.0 0.0.0.255 area 0
```

314P_076

R2 Configuration for Simple Password Authentication

```
<output omitted>
interface Loopback0
 ip address 10.2.2.2 255.255.255.0

<output omitted>
interface Serial0/0/1
 ip address 192.168.1.102 255.255.255.224
 ip ospf authentication
 ip ospf authentication-key plainpas

<output omitted>
router ospf 10
 log-adjacency-changes
 network 10.2.2.2 0.0.0.0 area 0
 network 192.168.1.0 0.0.0.255 area 0
```

Verifying Simple Password Authentication

```
R1#sh ip ospf neighbor
Neighbor ID     Pri   State           Dead Time   Address         Interface
10.2.2.2          0   FULL/  -        00:00:32    192.168.1.102   Serial0/0/1

R1#show ip route
<output omitted>
Gateway of last resort is not set
     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
O       10.2.2.2/32 [110/782] via 192.168.1.102, 00:01:17, Serial0/0/1
C       10.1.1.0/24 is directly connected, Loopback0
     192.168.1.0/27 is subnetted, 1 subnets
C       192.168.1.96 is directly connected, Serial0/0/1

R1#ping 10.2.2.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.2.2.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 28/29/32 ms
```

Configuring OSPF MD5 Authentication

Router(config-if)#

ip ospf message-digest-key *key-id* **md5** *key*

- **Assigns a key ID and key to be used with neighboring routers**

Router(config-if)#

ip ospf authentication [message-digest | null]

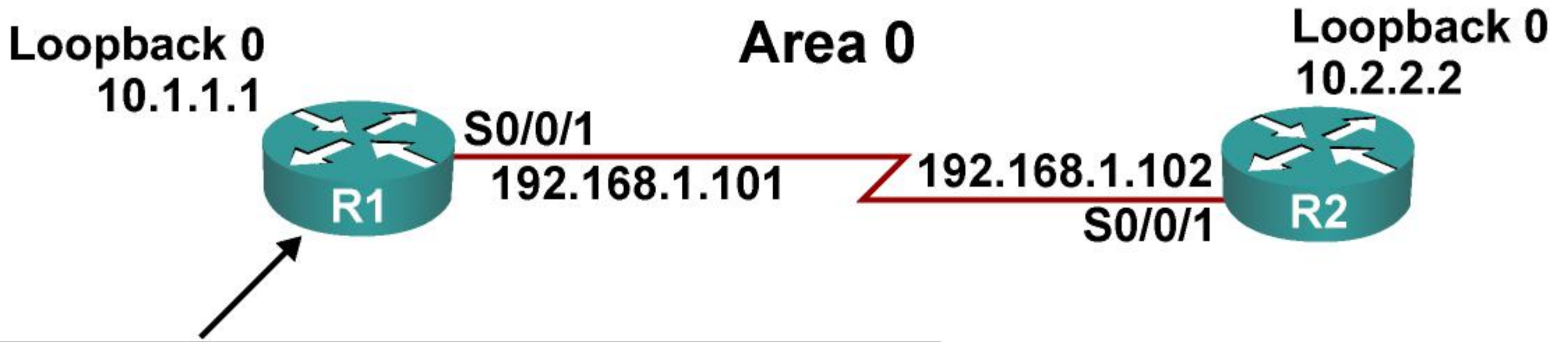
- **Specifies the authentication type for an interface (since Cisco IOS software 12.0)**

Router(config-router)#

area ***area-id*** **authentication [message-digest]**

- **Specifies the authentication type for an area (was in Cisco IOS software before 12.0)**

Example MD5 Authentication Configuration




```
<output omitted>
interface Loopback0
 ip address 10.1.1.1 255.255.255.0

<output omitted>
interface Serial0/0/1
 ip address 192.168.1.101 255.255.255.224
 ip ospf authentication message-digest
 ip ospf message-digest-key 1 md5 secretpass

<output omitted>
router ospf 10
 log-adjacency-changes
 network 10.1.1.1 0.0.0.0 area 0
 network 192.168.1.0 0.0.0.255 area 0
```

314P_077

R2 Configuration for MD5 Authentication

```
<output omitted>
interface Loopback0
 ip address 10.2.2.2 255.255.255.0

<output omitted>
interface Serial0/0/1
 ip address 192.168.1.102 255.255.255.224
 ip ospf authentication message-digest
 ip ospf message-digest-key 1 md5 secretpass

<output omitted>
router ospf 10
 log-adjacency-changes
 network 10.2.2.2 0.0.0.0 area 0
 network 192.168.1.0 0.0.0.255 area 0
```

Verifying MD5 Authentication

```
R1#sho ip ospf neighbor
Neighbor ID     Pri   State           Dead Time   Address         Interface
10.2.2.2          0   FULL/  -        00:00:31    192.168.1.102   Serial0/0/1

R1#show ip route
<output omitted>
Gateway of last resort is not set
     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
O       10.2.2.2/32 [110/782] via 192.168.1.102, 00:00:37, Serial0/0/1
C       10.1.1.0/24 is directly connected, Loopback0
     192.168.1.0/27 is subnetted, 1 subnets
C       192.168.1.96 is directly connected, Serial0/0/1

R1#ping 10.2.2.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.2.2.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 28/28/32 ms
```

Troubleshooting Simple Password Authentication

Router#

```
debug ip ospf adj
```

- **Displays the OSPF adjacency-related events**

```
R1#debug ip ospf adj
OSPF adjacency events debugging is on
R1#
<output omitted>
*Feb 17 18:42:01.250: OSPF: 2 Way Communication to 10.2.2.2 on Serial0/0/1,
state 2WAY
*Feb 17 18:42:01.250: OSPF: Send DBD to 10.2.2.2 on Serial0/0/1 seq 0x9B6 opt
0x52 flag 0x7 len 32
*Feb 17 18:42:01.262: OSPF: Rcv DBD from 10.2.2.2 on Serial0/0/1 seq 0x23ED
opt0x52 flag 0x7 len 32  mtu 1500 state EXSTART
*Feb 17 18:42:01.262: OSPF: NBR Negotiation Done. We are the SLAVE
*Feb 17 18:42:01.262: OSPF: Send DBD to 10.2.2.2 on Serial0/0/1 seq 0x23ED opt
0x52 flag 0x2 len 72
<output omitted>

R1#show ip ospf neighbor
Neighbor ID     Pri   State           Dead Time   Address         Interface
10.2.2.2          0   FULL/  -        00:00:34    192.168.1.102   Serial0/0/1
```

Troubleshooting Simple Password Authentication Problems

Simple authentication on R1, no authentication on R2

```
R1#
*Feb 17 18:51:31.242: OSPF: Rcv pkt from 192.168.1.102, Serial0/0/1 : Mismatch
Authentication type. Input packet specified type 0, we use type 1

R2#
*Feb 17 18:50:43.046: OSPF: Rcv pkt from 192.168.1.101, Serial0/0/1 : Mismatch
Authentication type. Input packet specified type 1, we use type 0
```

Simple authentication on R1 and R2, but different passwords

```
R1#
*Feb 17 18:54:01.238: OSPF: Rcv pkt from 192.168.1.102, Serial0/0/1 : Mismatch
Authentication Key - Clear Text

R2#
*Feb 17 18:53:13.050: OSPF: Rcv pkt from 192.168.1.101, Serial0/0/1 : Mismatch
Authentication Key - Clear Text
```

Troubleshooting MD5 Authentication

```
R1#debug ip ospf adj
OSPF adjacency events debugging is on
<output omitted>
*Feb 17 17:14:06.530: OSPF: Send with youngest Key 1
*Feb 17 17:14:06.546: OSPF: 2 Way Communication to 10.2.2.2 on Serial0/0/1, state
2WAY
*Feb 17 17:14:06.546: OSPF: Send DBD to 10.2.2.2 on Serial0/0/1 seq 0xB37 opt
0x52 flag 0x7 len 32
*Feb 17 17:14:06.546: OSPF: Send with youngest Key 1
*Feb 17 17:14:06.562: OSPF: Rcv DBD from 10.2.2.2 on Serial0/0/1 seq 0x32F opt
0x52 flag 0x7 len 32  mtu 1500 state EXSTART
*Feb 17 17:14:06.562: OSPF: NBR Negotiation Done. We are the SLAVE
*Feb 17 17:14:06.562: OSPF: Send DBD to 10.2.2.2 on Serial0/0/1 seq 0x32F opt
0x52 flag 0x2 len 72
*Feb 17 17:14:06.562: OSPF: Send with youngest Key 1
<output omitted>

R1#show ip ospf neighbor
Neighbor ID     Pri   State           Dead Time   Address         Interface
10.2.2.2          0   FULL/  -        00:00:35    192.168.1.102   Serial0/0/1
```

Troubleshooting MD5 Authentication Problems

MD5 authentication on both R1 and R2, but R1 has key 1 and R2 has key 2, both with the same passwords:

R1#
***Feb 17 17:56:16.530: OSPF: Send with youngest Key 1**
***Feb 17 17:56:26.502: OSPF: Rcv pkt from 192.168.1.102, Serial0/0/1 : Mismatch Authentication Key - No message digest key 2 on interface**
***Feb 17 17:56:26.530: OSPF: Send with youngest Key 1**

R2#
***Feb 17 17:55:28.226: OSPF: Send with youngest Key 2**
***Feb 17 17:55:28.286: OSPF: Rcv pkt from 192.168.1.101, Serial0/0/1 : Mismatch Authentication Key - No message digest key 1 on interface**
***Feb 17 17:55:38.226: OSPF: Send with youngest Key 2**

Summary

- When authentication is configured, the router generates and checks every OSPF packet and authenticates the source of each routing update packet that it receives. OSPF supports two types of authentication:
  - Simple password (or plain text) authentication: The router sends an OSPF packet and key.
  - MD5 authentication: The router generates a message digest, or hash, of the key, key ID, and message. The message digest is sent with the packet; the key is not sent.
- To configure simple password authentication, use the ip ospf authentication-key password command and the ip ospf authentication command.

Summary (Cont.)

- To configure MD5 authentication, use the ip ospf message-digest-key key-id md5 key command and the ip ospf authentication message-digest command.
- Use show ip ospf neighbor, show ip route, and debug ip ospf adj to verify and troubleshoot both types of authentication.
- With MD5 authentication, the debug ip ospf adj command output indicates the key ID sent.

OSPF Summary

- OSPF is an open-standard link-state routing protocol, offering quick convergence and the ability to scale large networks.
- There are five OSPF packet types: hello, DBD, LSU, LSR, and LSAck.
- Configuration of OSPF is a two-step process:
  - Enter OSPF configuration with the router ospf command.
  - Use the network command to describe which interfaces will run OSPF in which area.
- OSPF defines three types of networks: point-to-point, broadcast, and NBMA. On NBMA networks, OSPF mode options include nonbroadcast, broadcast, point-to-multipoint, point-to-multipoint nonbroadcast, and point-to-point.

Module Summary (Cont.)

- LSAs are the building blocks of the LSDB. There are 11 types of OSPF LSAs.
- Route summarization reduces OSPF LSA flooding and routing table size, which reduces memory and CPU utilization on routers.
- Stub area techniques improve OSPF performance by reducing the LSA flooding.
- OSPF supports two types of authentication:
  - Simple password (or plain text) authentication
  - MD5 authentication

Other Routing Protocols

Integrated System-Integrated System (IS-IS) Protocol

- IS-IS was defined in 1992 in the ISO/IEC recommendation 10589.
- IS-IS is a link-state routing protocol.
  - Provides fast convergence and excellent scalability. Very efficient in its use of network bandwidth.
  - Uses Dijkstra’s Shortest Path First algorithm (SPF).
- Types of packets
  - IS-IS Hello packet (IIH), Link State Packet (LSP), Partial Sequence Number Packet (PSNP) and Complete Sequence Number Packet (CSNP).
- Link States are called LSPs
  - Contain all information about one router adjacencies, connected IP prefixes, OSI end systems, area addresses, etc.
  - One LSP per router (plus fragments).
  - One LSP per LAN network.
- IS-IS has 2 layers of hierarchy
  - The backbone is called level-2.
  - Areas are called level-1.

Enhanced Interior Gateway Routing Protocol (EIGRP) Protocol

- EIGRP is a Cisco-proprietary protocol that combines the advantages of link-state and distance vector routing protocols.
- EIGRP has its roots as a distance vector routing protocol and is predictable in its behavior.
- What makes EIGRP an advanced distance vector protocol is the addition of several link- state features, such as dynamic neighbor discovery.
    - EIGRP Maintains a Neighbor Table, a Topology Table, and a Routing Table.
- EIGRP has Variable-length subnet masking (VLSM) support.
- Has a sophisticated metric that considers five criteria:
    - Two by default:
        - Bandwidth - The smallest (slowest) bandwidth between the source and destination.
        - Delay - The cumulative interface delay along the path.
    - Available, are not commonly used, because they typically result in frequent recalculation of the topology table:
        - Reliability - The worst reliability between the source and destination, based on keepalives.
        - Loading - The worst load on a link between the source and destination based on the packet rate and the interface's configured bandwidth.
        - Maximum transmission unit (MTU) - The smallest MTU in the path.
- A significant advantage of EIGRP (and IGRP) over other protocols is its support for unequal metric load balancing that allows administrators to better distribute traffic flow in their networks.

Fim

- Obrigado pela atenção.
- Questões?
- Comentários?